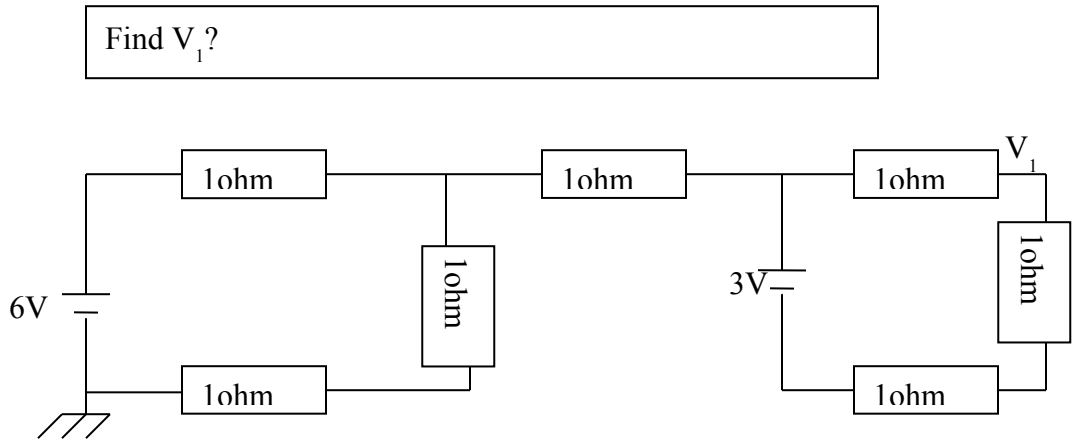
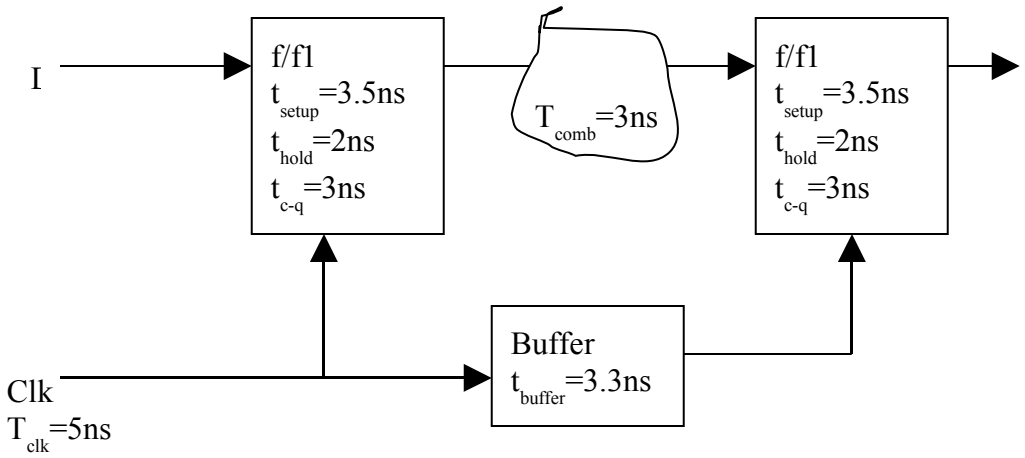
VLSI Interview questions

CISCO:

1)



2) Some star to delta conversion networks for finding the R
3) What shud we do to reduce latch up -----
3) How to reduce short channel effects – substrate is to heavily doped
4) Some mental ability q's
5) Convert a mux to an OR gate
6) Design a 2X1 mux using half adders
7) Some clk skew q
8) Some simple ckt which has 2 voltage sources in series to it and a current source u had to find the I through resistor which is a easy one to solve
9) Two latches constructed using muxes are cascaded such that it acts like a master slave flipflop and u shud mention wether it is +ve edge triggered or –ve edge triggered..
10) Some stuck at fault in a ckt and u shud mention the test vector for it.
11) Some k map simplification….
12) Given a boolean eq. and u shud design the ckt using min no. of nmos and pmos for that go for pseudo nmos technique.
13) Given the below ckt and u shud tell wether the clk period is enough or not and what problems that the ckt will faces (I m not able to remember the correct q and diagram)



Interview questions:
Some basic inverter q's
Latch up q's
Timing violation q's

Freescale:
1. How to design AND Gate using one pMOS and one nMOS.

2. Design a flip flop using MUX.

3.Design a divide by 3 synchronous circuit.

4. Positive edge detector circuit.

5. A simple combinational circuit was asked to be simplified.

6. Design a two bit comparator with and without using MUX.

7. A transistor circuit is given.find out the output voltage given Vbe and Vce. This is a simple one.

8. Design a square wave generator which takes only one positive edge trigger.

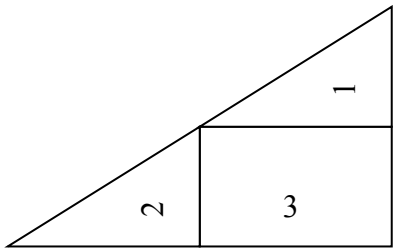
9. A question on maximum frequency of operation of a circuit. the setup time, hold time of the flip flops are given.

10. What is the purpose of the impedence matching between the load and source?
ans: To avoid the reflection of the power.

ITTIAM:

Written Test (Apti) :

1. Probability of 0 → 1 is p1 and 1 → 0 is p2. If 00 is xferred what is the prob of receiving at least one of them is 0. Ans 1 – (p1*p1)
2. Triangle 1: width is 5, height is 2.
   Triangle 2: width is 8, height is 3
   Rectangle 3: width is 5, height is 3
   What's the total area?
   a) 32
   b) 32.5
   c) 33
   d) both a and b



2. Something like this:

6471p + 3245q = 263452
3245p + 6471q = 236231

a) 1.5 <= p <= 2
b) 2 <= p <= 2.5 etc.

3. Speed downstream is 72 Kmph, level is 63 Kmph, upstream is 54 Kmph. A person travels A to B in 4 hrs and returns in 4 hrs 40 min. Distance from A to B is
   a) 203 Km
   b) 273 Km
   c) 302 Km
   d) Data insufficient
4. Something like this:
   A said "B didn't do it"
   B said "I didn't do it"
   C said "A did it"
   D said "B lies"
   Who is true?
5. Speed uphill=53miles/hr; speed downhill=70 miles/hr; speed on flat road=63miles/sec. It takes 4 hr to travel from town A to B and 4hrs 40min to travel from B to A. Find the distance between the towns.

6. A has n+1 coins and B has n coins. Both of them together toss all their coins. What is the probability that A gets more no of heads than B.

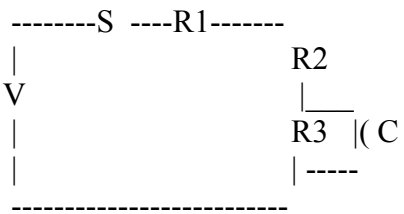
Interview

1. How to construct 4x1 mux using 2x1 mux only.
2. How to find out contents of PC at any point in the code. Ans Using CALL and reading top of stack.
3. x = (x +1) % 2 in the body of big loop. Optimize this to single operation. Initially x = 0. Ans x = not (x)
4. How to make a monostable (one shot) multivibrator using flipflops.
5. If the clock and D input of a D flipflop are shoted and clock connected to this circuit, how will it respond?
6. Some opamp circuit with several voltage and current sources connected through resistor dividers, find output.
7. Basic DSP theory: What is the frequency domain representation of (1) sinewave (2) cosine wave (3) the combination of sine and cosine waves. Given the output of (3)above, how will you find the input? Draw and show how it looks like.
8. If a LPF and HPF are connected in series, how will they respond under different cases of their cutoff frequencies (example if f1 < f2, what will happen)?
9. Interface an 8 bit µP with two 8Kx8 RAM chips. What would you do if A0, A1 are interchanged in h/w for only one memory chip. What'd you do in case of PROMs in case of RAMs?

DSP Paper

There are 3 sections ee, dsp and cse each with 20 q?s u have to attempt any one section only . Here I am sending the dsp section which I took , in other section the first and last 4 q?s were same as dsp.

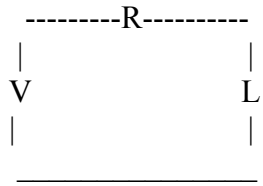
1



instantaneous Voltage across R2 when switch S is closed :
a.V*R2/(R1+R2)
b.V*R2/(R1+R2+R3)
c.0
d.V

Ans V*R2/(R1+R2+R3)------is what I wrote

2



as freq increases which of the following increases

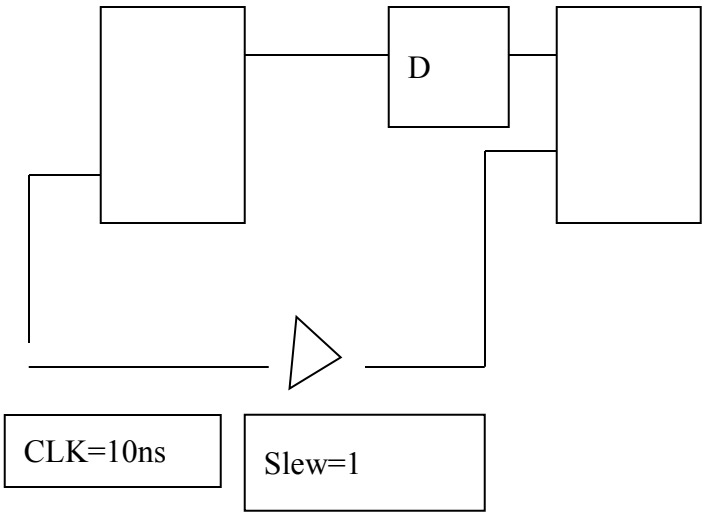
ans : Z and V(L)

3 q? on setup time and dealy

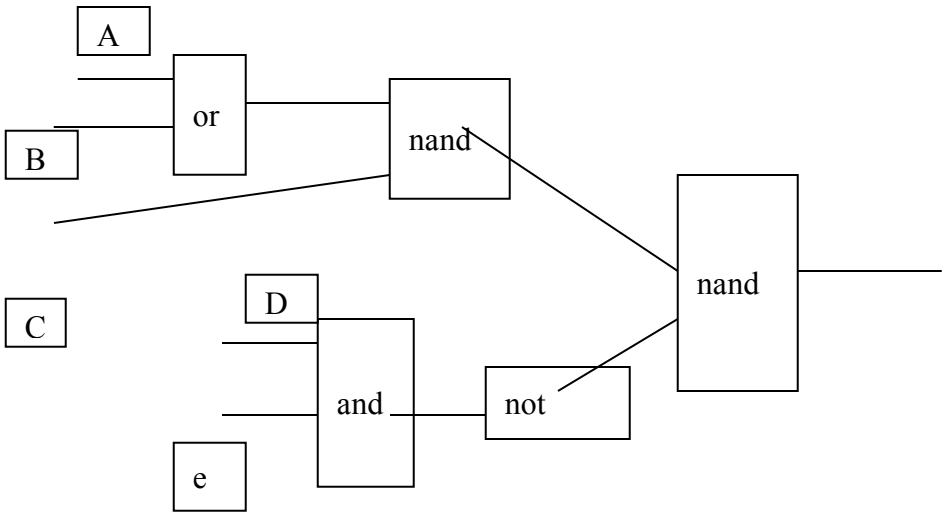
diagram below given not very clear\

clock period is 10ns ,
setuptime of each ff is 2 ns
clk2Q delay is 3ns
Slew dew to inverter is 1ns
Wht is the max allowable dealy of block D hold time=0

Ans 10-2-3-1=4ns



4 o/p of the following gate



ans : (a+b)c+de

5 SER=10^-4 the BER of a QPSK
a =SER
b <=SER
c>=SER
d =SER/2
ans >=SER

6 for 62db of PCM System what is the no of bits =10

7 for a 4 level pipeline processor the no of machine cycles required for executing 4 and (someno I don't rember) with initially pipeline flushed

ans = 4+3 and …+3
u add there for initial latency

8 An ideal LPF is
a causal
b non causal
c non stable
d none
ans: non causal ,

9 impluse func and white noise have same
a *magnitude and phase response*
b magnitude response
c phase response
d none
ans magnitude and phase response

10
y(t)=y(t-1)+0.1x(n) is what typr of filter
ans : IIR LPF

11 a signal s(t)=sin(omega*t) is sampled at fs, then the resulting signal spectrum is periodic depends on:
a *Omega/fs*
b omega *fs
c omega
d fs
ans omega/fs

12 if 2 gaussian func of mean m1 and m2 are added the wht is the resulting PDF
a guassian func with mean m1+m2
b guassian func with mean m1+m2/2
c uniform with mean m1+m2
d rayelig with mean m1+m2

13 u(t)+ sumof(deltafunc(n-k)){n=- infinitive to+ infinitive) is equaltent to
I hope I am made this clear it is a simple one
Ans u(-t-1) +2u(t)

14 if the probable of drawing an even no is p the wht is the probailty of drawing odd no in 2nd chance given 1st draw resulted in even one
a.p
b.(1-p)
c.p(1-p)
d.p/(1-p)
ans p(1-p)

15 no of multi required to mutli 2 upper triangular matrixes
a p(p+1)/2
b (p-1)(p+1)/2
c.summation (i=1 to p) i(i+1)/2
d…..
ans:c

16 a c program given something like this
unsigned short int i, j=1;
for (i=0,i<10,i++)
if(i&j)
printf("ITTIAM");

how many time is Ittiam printed .
ans 5 since bitwise and therefore only odd no will result in true if condtion

17 some c program abt function concerned with pointer and local variable easy one
ans 25

18 f=100 khz fs=125khz
o/p of filter with cutoff 150khzs
ans 25 and 100 khzs

19 stack in a processor is used for
a function call
b unlimted function call
c local variable
d something
I am not very sure if it is function call or unlimited function call , since function call also be done with shadow registers but only to a certain depth but most processor don't use shadow registers . at the same time depth of stack is also limited

20 x(F) is a signal whose freq response is asymmetric i.e H(F)=H(-f)
then it can be concluded that x(t) is
a real
b real or complex
c comples
d none
ans complex

here is the apti paper it also has 20 q?s and 30 min

1 some q? on some no is appended with 7(on right of units place) multiplied by 5 then result is similar to intial no with 7 on the left most (most significant digit)find the 3rd digit
ans =2 (?)

2 OTTSSFF?N
a.T
b.E
c.N
d…
ans: I wrote E

3 abcdefghij
a=no of zeros in the no
b= no of ones in the no

c= no of twos
so on
wht is the sum of digts
a.10
b.55
c.9
d….
ans 9(?)

4 ¼ of a no +2/3 of another no =3/8 of sum
wht is their ratio
ans: 3:7 (7:3)

5,6,7,8 9 four q? on some gre type analytical it was abt some 4 family runs a 4 restaurant name of husband (jai, jayesh, Parikh,bipin), wife(beena, chand, preethi, sangeetha) and their familyname (joshi,natwar,sahni,….)give , some hint and who runs which hotel(Indian court, American court,….) asked

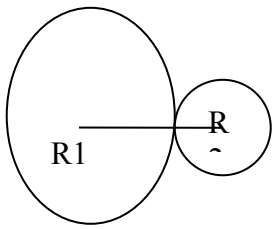
ans D D D C in that order in our paper :ANSWER THIS Q (I think this was an important question)

10 If in a test 1 mark is for correct answer ,what negative mark should be kept for nullifying the correct answers…
a.1/4
b.1/2
c.3/4
d…
ans=1/4

11 A wins B by 28 meters or (some) seconds(time) the A is ahead of B

Ans 4 min 20 sec

12 Given 2 circles of radius R1 & R2. how many rotations will the smaller circle have to make a full revolution around the circle with radius R1.



ans (r1+r2)/r1

13 An equlateral triangle and its circumcircle..what is the probability that a line drawn inside this circle is longer than the side of the equilateral triangle

ans =1/3(?)

14 given wt 1,3,9,27 how much max can u weigh
ans 40

15 43 players play some knock out game . how many games should be conducted to declare a winner
ans 42

15A man traveling at a speed of….misses a train by 7 mins…if he travels at a speed …..how far should he travel to catch the train
ans 6 km

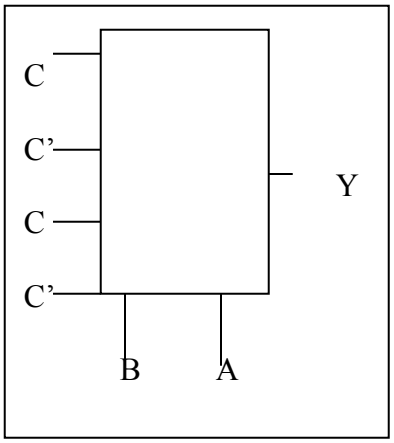
16 2 traingle made form circles fiven

0	0000
00	000
000	00
0000	0

how many min circles have to be removed to get some thing
ans 3

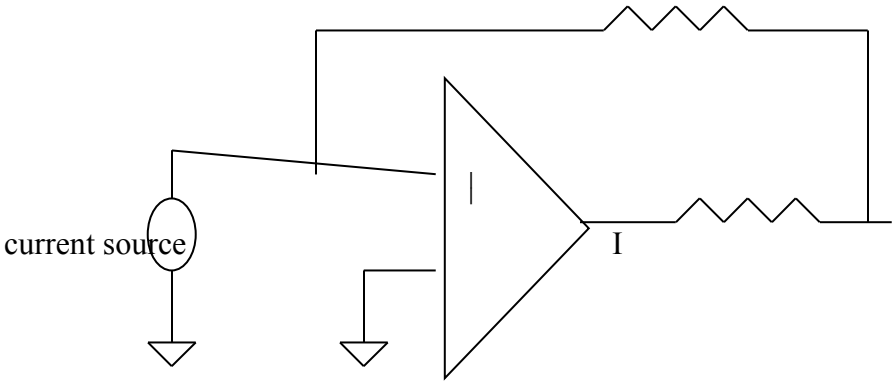
17 a boy has trasfered 100 galss from one palce to another the owner puts a condition tht if he delivers safely he gets 3 paise for each glassand he would forfeit 9 paise for every broken glass. He loses some glass and gets Rs.2.40/- wht are the no of broken glass
ans 5

in EE paper:
some questions which I saw:
5. what is the expression for o/p?
a.AC
b.A'C+AC'
c.C
d.B'C+BC'

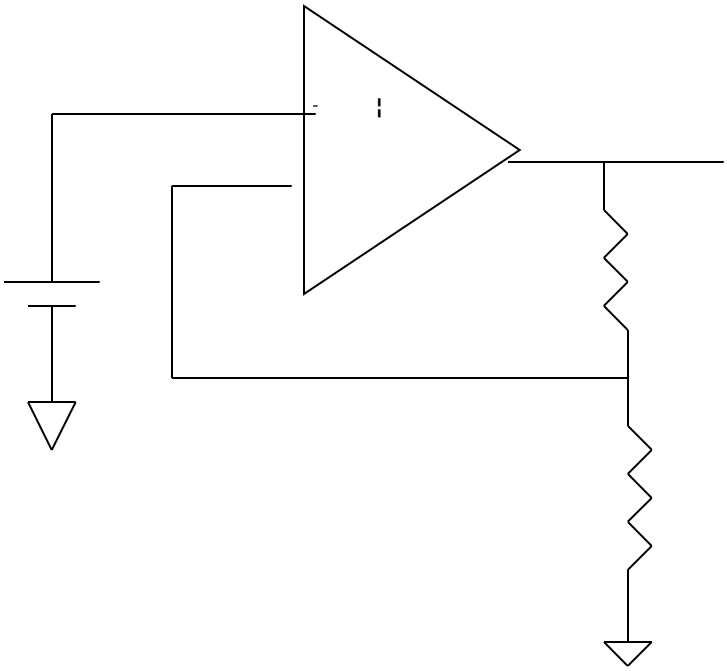


ans:A'C+AC'

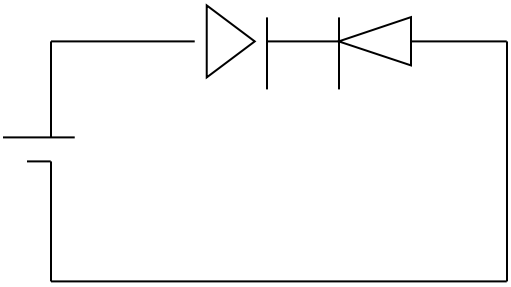
6.what is the current I in the ckt assuming ideal opamp as shown: resistance values given



7.what does the ckt below work as:
something similar to this : I think it was Schmitt trigger ckt (check it out)



8.what is the current flowing in the ckt:
a.Is
b.0
c…
d….



9.some problem on writing laplace transform of the given ckt
a.E/s[….]
b.Es[…]
c….
d….

Hi …… These are the questions,in interview for ITTIAM. I have written the answers I gave. There might be better solns ☺

Q. Give the time and frequency domain representation of a sine wave.
A. Sketch a sine wave for time domain rep. For freq domain rep, it is 1/2j [delta(f-fc)-delta(f+fc)]. Phase of the sine wave is 90 deg. (lag)

Q. Repeat the above for a cosine wave – Phase is zero deg.

Q. Repeat the above for sum of sine and cos waves.

Q. Given large amount of data of ones and zeros, how wud u compress it using huffman coding.
A. Use Run Length coding and code the run lengths using Huffman coding

Q. Any alternate way for the abov prob.
A. Consider three bits at a time. That can give rise to eight possible symbols. Then code these symbols using Huffman Coding.

Q. Will Huffman coding be always advantageous.
A. No, If the prob of occurance of symbols are equal then there is no advantage

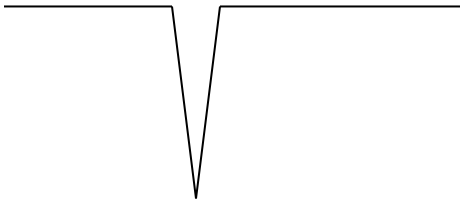
Q. When is Huffman coding optimum
A. When prob of symbols are powers of ½ it can be optimum

Q. why
A. Use the equation, bits reqd = – log p. If p is powers of ½ then no of bits will be an integer and hence it will be optimum.

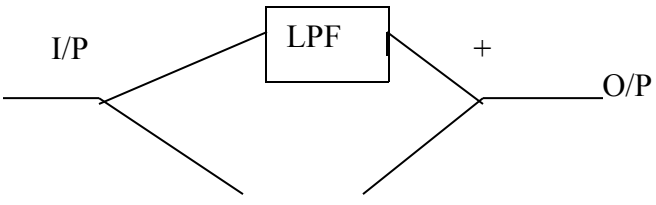
Q. Is there any other method of optimal entropy coding
A. Arithmetic Coding

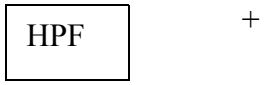
Q. Why is it optimal
A. Since group of symbols are coded, fractional bit rate can be used.

Q. What is a notch filter
A. Filter with response



Q. Given Low pass and High pass filters how to realise above filter
A.





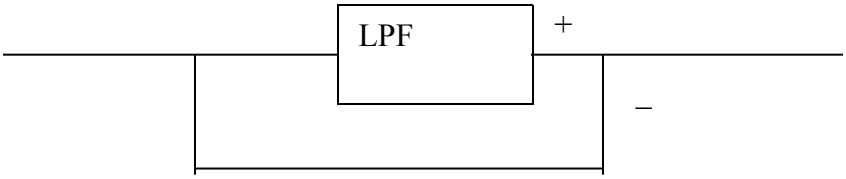
Choose LPF with cut off slightly below the notch freq.
HPF with cut off slightly above the notch freq.

Q. How to design filter with gain at single freq
A. Put LPF and HPF in series. Choose Cut off of LPF slightly above the notch freq.
Choose cut off of HPF slightly below notch freq.

Q. Given an RC-Ckt, Low pass output is obtained across which component and why
A. Across C. Because impedance of C inc as freq reduces, so voltage across it inc, as freq reduces.

Q. Given two LTI systems in cascade what is the resultant gain and phase
A. Resultant Gain is product of the two gains, Resultant Phase is sum of the two phases.

Q. what is the O/P of the following system



A. All the low freq components will be attenuated and high freq components will be inverted

Q. How will the phase of LPF affect the above ckt.
A. Low frequencies get cancelled only when phase shift is integer multiples of 2pi.

Q. Two LTI sys are in cascade. Impulse response of first system is h1(n). Second system is described by the diff equ. Y(n)=x(n-1)+x(n), what is the overall response of the cascade
A. overall response is h1(n-1) + h1(n)

Q. I have a database of 5 faces. Given a test image of some face, what is the simplest way to recognize it.
A. Correlation

Q. If the test face is taken in totally diff lighting conditions wud correlation work, How wud o solve the prob
A. I gave solns like removal of DC Component, Histogram equalisation and then correlation etc…….he was not convinced.

Q. Is an image zero mean signal
A. No

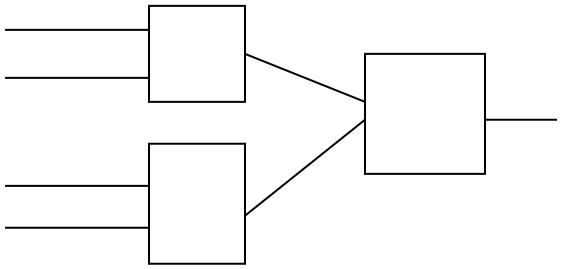
Q. P.T. sum of 2 odd nos is even

Q. Is sum of two prime nos prime. If Yes Prove, If not Prove

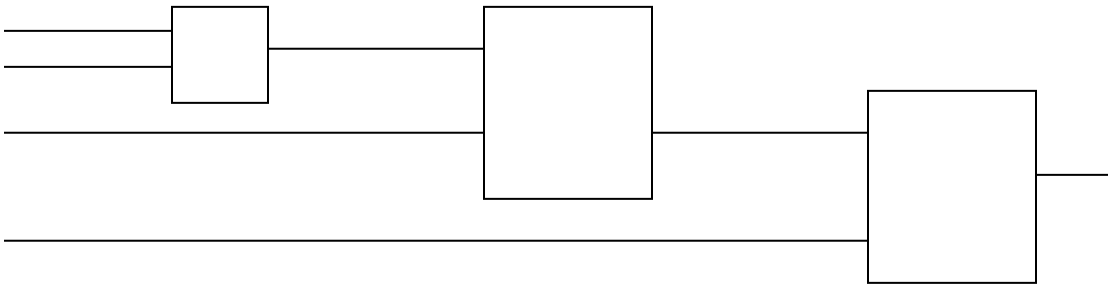
A. Same proof for both of the above

Q. Give diff configurations to realise 4 - I/P AND gate using 2 – I/P AND gates
A.

Config 1:



Config 2:



Q. Given a black box consisting of one of the above config, how wud u detect as to which is the config.
A. In Config 1, delay is same for all i/ps. In Config 2 delay for i/p 1110 is less than for 1011.

There were other small questions which I do not remember. All were very very basic. Just stay cool and u can answer everything.

ITTIAM paper 2004 (EE section only)

1) If the probability of 0 being received as 1 is p_1 and that of 1 being received as 0 is p_2. What is the prob of receiving at least one 1 if two consecutive zeroes are sent?
→$1-(1-p_1)^2$

2) For symbols a, r, and p having a prob of occurrence as 0.4, 0.2, and 0.2 and t being another symbol, what is the length of code for 'a' in huffman binary coding?
→ 1

3) For probability of 0.25,0.25,0.5, find the entropy.
→3/2

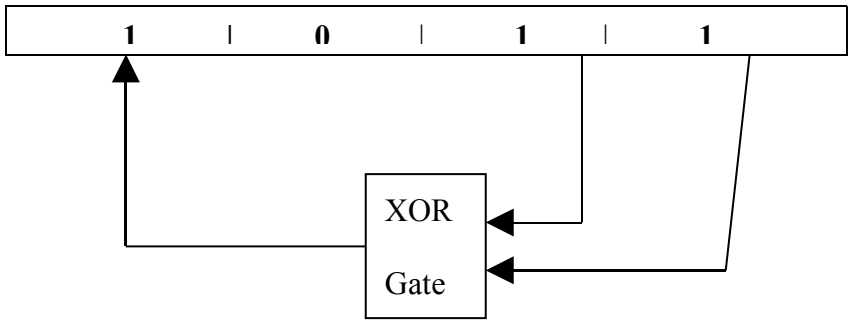
4) How many multiplications will be required for multiplying two p×p upper triangular matrices.
→ sigma p^3 (I am not sure of the answer)

5) How many cycles are required for a N,M convolution given that each addition, mult and mac requires one cycle?
→ I ticked (N-M+1)M/2 (not sure of the answer)

6) For 62db of SNR, what it the channel capacity? (think they meant per unit BW)
→ use $C=B \log_2(1+SNR)$

7) One question on odd-parity detector. ('what is this' type)

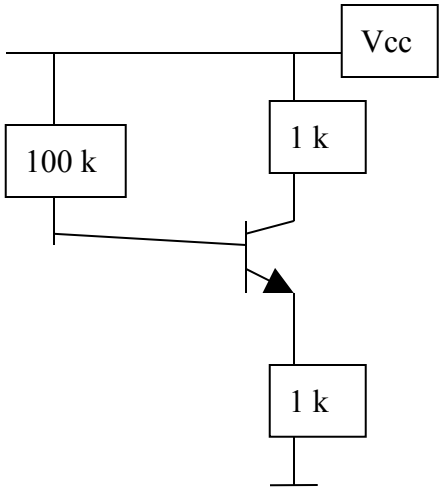
8) One question on bistable multivibrator using 741. (again 'what it this' type)



9) The above is a 4 bit shift reg. The feedback path has an XOR gate. Tell the value of the reg after two shift right.

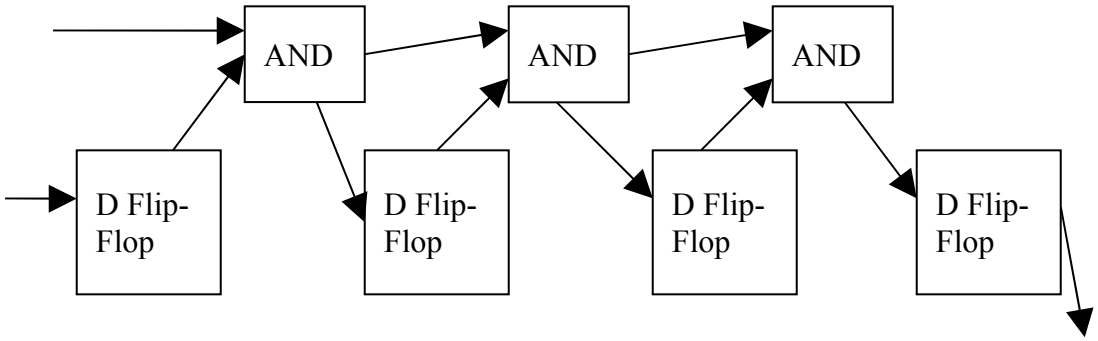
10) A simple C program. What will be the output of i&j where i=10, j=20.
→ 0

11) For Vcc =20 V and beta =100 find Ie for V_{be}= 0.6 V. Take Ic = Ie.
→9.6 mA



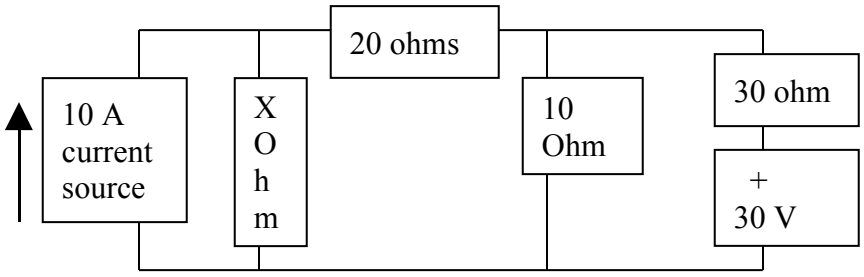
12) The propagation delay of each AND gate is 10 ns. What could be the max clock frequency.

→ 10^8 Hz

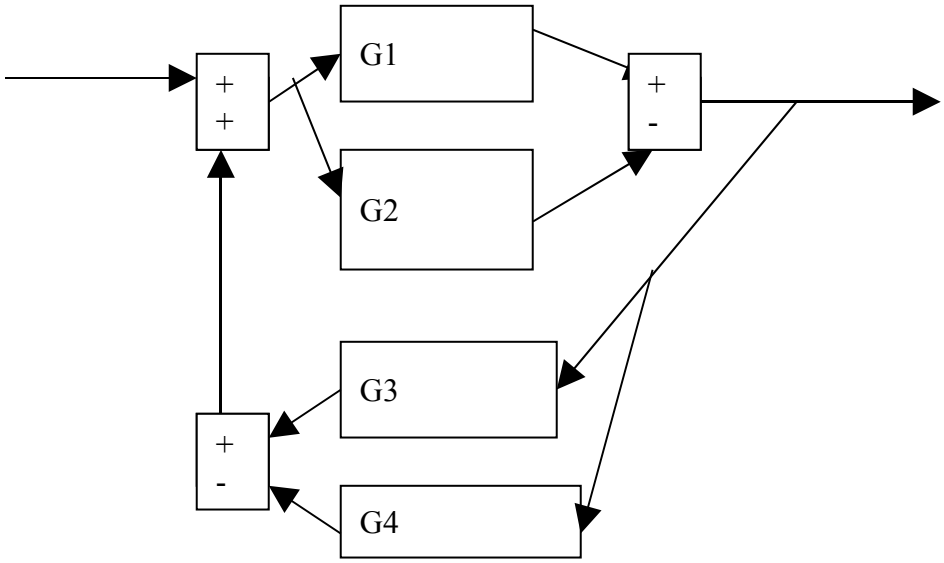


13) The current across 20 ohm res is

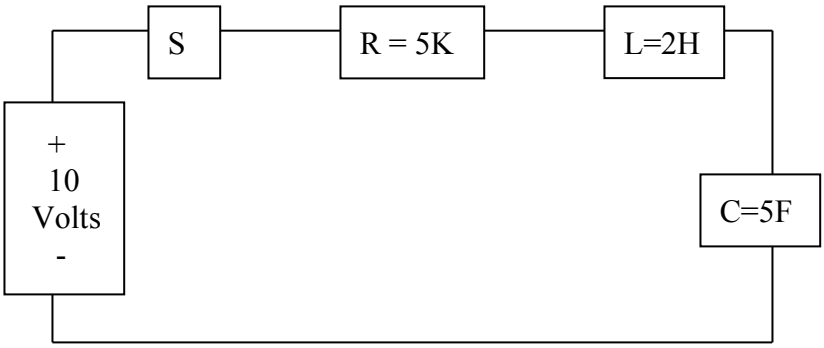
→0.5



14) Block diagram (find the Transfer Function):



15) Find I for t=0 when the switch S is closed. Also dI/dt.
→ 0A, 5A/s



16) There was a question about the causality and non-linearity of a system given its difference equation.

Ittiam interveiw questions

1. make a 4x1 mux using 2x1 multiplexers
2. make a 2-input Or gate using 2x1 mux
ans. make A & B as the input of Mux & B as the select signal
3. In the z-plane there is a zero at intersection of unit circle and x-axis & there are two poles somewhere inside the unit circle Then what can u tell about the fourier transform of the signal??
ans. the FT will definitely be zero at origin
4. A signal s1(t) is passed through a LPF to get s2(t) and the s2(t) is

subtracted from s1(t) to get s3(t) =s1(t)-s2(t)
How will s3(t) relate with s1(t)??
ans. It may or may not be the high pass filtered version of s1(t)
depending on the phase of LPF (he was not convinced with the answer)
5. what is difference between A-law & mu-law ??
ans. ??

Some questions of Ittiam:

'ABCDEFGHIJ' is a number.
A is the number of 0s.
B is the number of 1s etc
J is the number of 9s
What is the number

Some question on logic was there. like finding whose husband is john or whose wife is mary type.

in general other aptitude questions were peace

(interview) what is the algorithm to find if p is a prime. Why do u need to test only upto sqrt(p) factors.

Technical test and interview):
DSP: y(n) = a y(n-1) + b x(n); what is the condition for system to be stable ? why?
find region of convergence of 1 +2 z inverse
if y1(n)= h1(n)*x(n) and y(n)= ay1(n)+by1(n-1). Find impulse response of over all sysemin terms of h1(n) only.

Huffman coding.(given set of symbols and probabilities construct code)
ASked the algorithm in general to generate such an optimal code.(there is something like a tree diagram given in every digi compression book).

asked to give output of RC circuit with const voltage source. then with const current source. Asked why in 2nd case the voltage cannot goto infinity(could be capacitor break down or that the voltage across current source can't exceed a certain value)

Construct four input and gate using 3 two input and gates in two ways.
If a black box with one of them is given how will u find which configuration it is. U have only black box and nothing to compare that with. u can give any input and see output.
How many min number of input combinations do u need

Pseudo code for matrix transpose. Should be optimal and swapping alos should be optimal. I gave sswapping using arithmetic operators. But they wanted that using logical operators. jus replace the arithmetic operations with xor. how many computations do u need for the getting the transpose of n*n matrix. Why

Whats the probability that u pick two red balls out of a bag of two red balls and 3 black balls? they tried to confuse. But i held on . They were seeing it in a different manner but finally landed up with the same answer i gave. Soif ur sure don't give up

Intel:

Paper I

1. Find Voltage across R and C in the following circuits.

a. In a given RC circuit find the voltage across C and R?

b. In a given CR circuit find the voltage across R and C ?;

2. For the given _expression Y=A'B'C+A'BC+AB'C+ABC+ABC' realize using the following

a. 2 input and 3input NAND gate

b. 2 input and 3 input NOR gate

c. AND,OR, INVERTER.

d. INVERTER;

3. What is the importance of scan in digital system.;

4.Given A XOR B =C, such that prove the following

a. B XOR C =A

b. A XOR BXOR C=0;

5. Construct an input test pattern that can detect the result E stuck at 1 in the ckt below

NAND (A,B)->E, NAND(C,D)->F

AND(E,F)->A.

6. In a given opamp ckt input offcet is 5mv,volatage gain =10,000,vsat=+-15v

such that find the output voltage .

7. Draw the p side equation of the circuit.(I am not sulre about it)

8. Make a JK FF using a D FF and 4->1 MUX.

9.Use 2->1 MUX to implement the following _expression

Y=A+BC'+BC(A+B).

10.For the following ckt what is the relation between fin and fout.?

the D FF use +ve edge triggered and have a intial value is 0

CLK->two DFFs with complementing (i.e one DFF have CLK and other one have

Complement of it),inputs of DFF is same and output of DFFs is given to NOR

Gate and output of NOR gate is feedback to the two DFFs.

11. Design a asyncronous circuit for the following clk waveforms.

CLK->thrice the CLK period->half the period of input.

12. What is the setup time and hold time parameters of the FF, what happens if we are not consider it in designing the digital ckt.

13. Given two DFF A,B ones output is the input of other and have the common clock.

Fmax if A and B are +ve edge triggered, if A is+ve edge triggered ,B is -ve edge triggered what is the Fmax relation to previous Fmax relation…

14. What are the FIFOS .? give some use of FIFOS in design.

Paper II

1. What is FIFO ? where it is used?
2. what is set-up and hold time?
3. Two +ive triggered FFs are connected in series and if the maximum frequency that can operate this circuit is Fmax. Now assume other circuit that has +ive trigger FF followed by –ive trigger FF than what would be maximum frequency in terms of the Fmax that the circuit can work?
4. layout of gates were shown and u have to identify the gates (NAND & NOR gates)
5. make a JK FF using a mux(4:1) and a FF.

6. the waveform of clk, i/p and o/p were shown and u have to make a seqential circuit that should satisfy the required waveform.
7. resistor is connected in series with capacitor and the input is dc voltage. Draw the waveform across the capacitor and resistor.
8. two FFs, one is –ive triggered and other is +ive triggered are connected in parallel. The 2 i/p NAND gate is has the i/ps from the q_out of both the FFs and the output of the NAND gate is connected with the I/p of both FFs . Find the frequency of the output of the NAND gate w.r.t clk.

Interview questions (face to face discussion)

1. Draw the circuit for inverter. How does it work?
2. If the pmos and nmos is changed in the inveretr, how does it behave?
3. Design flow for ASICs and FPGA. what are the difference between the ASICs and FPGA?where do u use ASIC and where u use FPGA?
4. What is floorplanning?
5. What do u mean by technology file used in the synthesis or optimization for the circuit (netlist)? What is the difference in the technology files used for the ASICs and FPGAs based designing?
6. Using a FF and gates. Make a memory (i.e include RD, WR etc.)
7. If the setup & hold time gets violated than what u 'll do to remove it?
8. What is clock skew? How u 'll minimize it?
9. What is clock tree? How it looks like? Concept behind that.
10. What about the Vdd and Gnd lines ? does one Vdd and Gnd pins will be sufficient for the chip. What will be the effect of using single Vdd and Gnd pins in the chip?
11. What is voltage refernce circuit? What is bandgap? How does it work?
12. what is FIFO? How does it work? Draw the circuit of FIFO of 1-bit and 4memory location deep? What would happen if memory is full and again u try to write in FIFO? What u 'll do to overcome this problem? Which one would be more easier to implement :- either dropping the packet, when the FIFO is full or pushing the data of FIFO every time. And why ?

The following questions are used for screening the candidates during the prescreening interview. The questions apply mostly to fresh college grads pursuing an engineering career at Intel.

COMPUTER ARCHITECTURE QUESTIONS

1. For a single computer processor computer system, what is the purpose of a processor cache and describe its operation?

2. Explain the operation considering a two processor computer system with a cache for each processor.

What are the main issues associated with multiprocessor caches and how might you solve it?

3. Explain the difference between write through and write back cache.

4. Are you familiar with the term MESI?

5. Are you familiar with the term snooping?

STATE MACHINE QUESTIONS

1. Describe a finite state machine that will detect three consecutive coin tosses (of one coin) that results in heads.

2. In what cases do you need to double clock a signal before presenting it to a synchronous state machine?

3. Design a FSM that will assert output when more than one ‘1’ is recieved in last three samples. Do not use more then 4 states.

SIGNAL LINE QUESTIONS

1. You have a driver that drives a long signal & connects to an input device. At the input device there is either overshoot,

undershoot or signal threshold violations, what can be done to correct this problem?

VALIDATION QUESTIONS:

What are the total number of lines written in C/C++? What is the most complicated/valuable program written in C/C++?

What compiler was used?

Have you studied busses? What types?

Have you studied pipelining? List the 5 stages of a 5 stage pipeline. Assuming 1 clock per stage, what is the latency of an

instruction in a 5 stage machine? What is the throughput of this machine ?

How many bit combinations are there in a byte?

What is the difference between = and == in C?

Are you familiar with VHDL and/or Verilog?

MEMORY, I/O, CLOCK AND POWER QUESTIONS

1. What types of CMOS memories have you designed? What were their size? Speed? Configuration Process technology?

2. What work have you done on full chip Clock and Power distribution? What process technology and budgets were used?

3. What types of I/O have you designed? What were their size? Speed? Configuration? Voltage requirements?

Process technology? What package was used and how did you model the package/system?

What parasitic effects were considered?

4. What types of high speed CMOS circuits have you designed?

5. What transistor level design tools are you proficient with? What types of designs were they used on?

6. What products have you designed which have entered high volume production?

What was your role in the silicon evaluation/product ramp? What tools did you use?

7. If not into production, how far did you follow the design and why did not you see it into production?

nVIDIA:

```
1-6 are multiple choice questions)

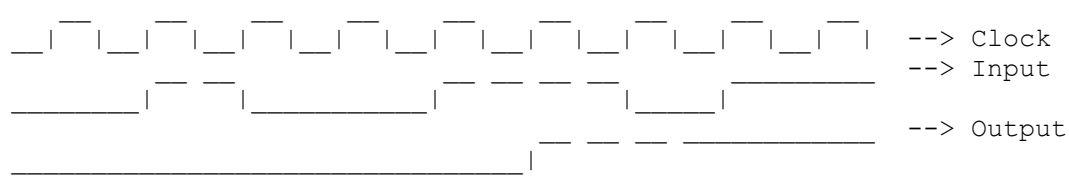
1. The max. value and min. value of 16-bit 2's complement (hex, dec,
binary)?
2. The max.,min. value of 16-bit 1's complement (hex, dec, binary)?
3. max. no. of logic functions for n-variables? ans: 2^2^n
4. about physical and virtual address, which is greater?
5. TLB (Translation Lookahead Buffer) is used for ?
   options:
   1. L1 cache misses
   2. L2 cache miss
   3. some thing page miss (not remembered exactly)

6. Minimum no. of P and N Mos transistors required to implement The
logic
   Function Y= !(A | B & C) using CMOS
   1. 1 p & 3 n
   2. 3 p & 1 n
```

3. 3 p & 3 n

7.One shot digital circuit

8.Noise eliminator (both -ve and +ve pulses of one clock cycle duration)



Observe the one clock delay in output.

Design a state machine for this operation.

9. Implement the following 2 functions using only 2x1 MUX without gates. U can use 1 or 0 for inputs.

a) Y = AB+not(C) b) Y = A xor B

10. Write a program in C or C++ to implement Stack and its functions such as
isEmpty, isFull, Push, Pop, Init etc. (I think based on the question given,
array implementation is sufficient)

11. Question on clock frequency required for given Sequential ckt.
Given 2 F/Fs, 3 delays (2 delays for clock dly1, dly2 and 1 delay dly3 of
combinational circuit) given setup, hold and propagation times

Ans: T+(dly2-dly1) >= Tpd + dly3 + Tsetup
Tpd + dly3 > Thold + (dly2-dly1)

Others:

1. A Positive logic NAND gate will be equuivalent to a '-'ive logic ---------- gate.

a) NAND
b) EX-NOR
c) NOR
d) OR

SANDISK:

SANDISK IIT BOMBAY PAPER, 26th DECEMBER, 2005

Written Test 45 mins

1) No. of universal logic gates reqd to implement EXOR

a) 4 NAND
b) 4 NOR
c) 5 NAND
d) 5 NOR

2) Using (A AND Bbar), we can implement
   a) only AND
   b) only OR
   c) any logic function
   d) none

3) A –V to +V pulse voltage source is connected to a RC series ckt. Draw the waveforms of voltage across R, voltage across C, and current in the circuit.

4) Draw the capacitance vs voltage characteristics of MOSFET and MOS cap, and point their differences in the HF region

5) Arrange an underdamped, critically damped, and overdamped system in order of phase margins

6) Find the voltage gain of a transconductance amplifier of transconducatnce gm, with Vi at +ve terminal, C btwn –ve term and gnd, and R between –ve term and output

7) Considering MOS caps Cgs and Cgd,
   a) Cgs>Cgd in cut-off region
   b) Cgd>Cgs in saturation region
   c) Cgd=Cgs in triode region
   d) None

8) Draw the waveform of "A" from the verilog code

```
Always(@clk)
Begin
A=0;
#5 A=1;
end;
```

9) Draw a NORbased latch, calculate its setup time if delay of each gate is td

10) A 1V dc source is connected to the source of an NMOS, a 0.1 nf cap is connected to the drain, and a 5V single pulse of duration 1 us is applied to the gate. To act as an integrator,
   a) W/L >>1
   b) W/L<<1

c) W/L=1
d) Cant be said from the given data

Interview 1st round :

Questions from the written test which I could not answer correctly, transfer characteristics of a CMOS inverter, implementation of an FSM given a state diagram, and a riddle :-given only a 3 l and a 5 l bottle, and nothing else, how would u measure 4 l water?

Interview 2nd round :

What are the issues if the duty cycle of the clock in a digital ckt is changed from 50%?
What are the different tests you would do to verify your verilog code?
How would your friends describe you?
What is the greatest risk you have taken so far in life?
What are the differences between academics and industry?

Paper II
1 simple current mirror question.

2 to generate non-overlapping clock.(see Rabaey page 339)

3 question on Verilog synthesis

4 always@(posedge clk)
begin
a=0;
#5 a=1;
end
what is the output waveform of a?

5 question on differential amplifier gain with $(w/l)_1=2*(w/l)_2$

6 V=vin1 – vin2 (vin1 and vin2 are two input voltages of 1 stage diff. amplifier).Now V is varied from -5 to +5 then draw the output voltage vs V.(Vdd=+5 Vss= -5).

7 two simple question on charging of capacitor with constant current source.

8 draw the VTC of buffer (PMOS and NMOS are interchanged in inverter)

9 what should be the ratio of $(W/L)_{PMOS}$ / $(W/L)_{NMOS}$ for switching threshold of Vdd/2.Given Kn/Kp=2.8.

10 there is 2 input CMOS NAND gate .inputs A and B changes from 0 to Vdd. but A goes to Vdd after B(after some delay). which input should be closer to Vout.

11 what are the benefits of finger layout----less junction capacitance etc

Others couldn't recall ..

Two rounds of interview-- HR and technical.
Nearly 45 mins for technical (Device, digital and mostly analog).
HR also of 45 mins.

Paper III

Q1) why noise margin in invertor calculated when slope becomes -1

Q2) one question on OTA acting as HPF (resistance with -ve f/b) and a capacitance at vin-
ans: gm(1+rsc)/gm+sc

Q3) question on verilog synthesis

Q4) draw c-v w/f for mos capacitance and mosfet

Q5) an ideal current pulse source charging a capacitance what wud be voltage across it

Q6) 3 step response given wat wud be the relative phase margin

ST Micro:

There were two papers
1. separate for the hardware (electonics people(VLSI))
2. and other for the embedded software design(both for electronics and comp).

1. which conversion is not possible
   a. float to int
   b. int to float
   c. char to float
   d. all are possible

2. threads have which thing in common
   a. register set
   b. data section
   c. thread id
   d. ...

3. one que like

```
main()
{
int x=5, y;
y= x*x++ * ++x ;

// print x and y
}
```

4. A CPU has four group of instruction set A, B, C, D
   CPI of A = 1
   CPI of B=3
   Cpi of c =2
   cpi of d= 4
   the cpu access 20% of A, 30% of b, 30% of C and 20 % of D
   what will be the average CPI.

   (this que was repeated in section 2 & 3)

5 . a question on hit ratio n effective memory access time.

6.

```
main()
{
int a=10,b=5
while ( --b>=0 && ++a)
{
--b;
++a;
}
print (a);
print (b);
}
```

7.

```
main()
{
char i;

for (i=0; i<=255; i++)
{
printf("%c", i);
}
}
```

8. One question on controls systems
   to find the transfer function.

poles n zeroes were given in a graph

9. One question on sampling theorem,
highest frequency of a signal is f, and it is sampled at fs(>= 2f). what is the frequency range of a bandpass signal whose spectrum looks exactly like the original signal.

10.One on the signal to noise ratio
There are N bits to represent each sample and total number of levels is M. If the amplitude range is reduced by half then SNR will be reduced by:

11. A cellular network operator is operating at 9.6Kbps. If he want to transmitt the audio quality of 44.1KHz with 16 bits for each sample how much bandwidth should be increased.

12. calculating the checksum for the bits to be transmitted given the frame-11000101 and generator is1100.

13. calculating the no of bits required for the error detection & the error correction for the given codeword set.
codeword a:
0000
0001
0011
1111
codeword b:
101111
.
.
.
.
110101

14. options were given to choose as which was an example of multitasking.
a:multiple remote users accessing a server
b:user working on spreadsheet, downloading some matter from internet
c:multiple programs resident in memory

15. CA in CSMA/ CA stands for
a. collision approval
b. collision avoidance
c. critical access

16. in a triangle, without changing the angle, if we double the sides,then new area will be

17. there is a pipe having dia 6mm, then how many pipes having 1mm dia wiill be needed to provide same amount of water.

18. in which of the folwng schemes after page replacement the entered page will enter in the same memory location as of the replaced one

a. direct mapping
b. n-set associative
c. associative
d. none of them

19. belady anamoly is related to.
ans. page replacement algos

20.which one uses cache mechanism
ans TLB

21.what will happen in following code..
signal(mutex)
critical section
wait(mutex)

ans. violation of mutual exclusion

22.an RLC ckt was given, fuctioning of ckt to be determined.
a: will act like FM
b: PM
c:AM
d: none of the above

23.

```
int i=0;
switch(i)
{
case 1: printf("hi");
case 0: printf("zero");
case 2: printf("world");
}
```

24.which one is the declaration of static string
a: static string
b: 'static string'
c: "static string"
d:char sting[30]

25.a que on file handling in c
a: file cant be opened
b:msg.txt is copied to msg
c:only first string be copied
d:

26. which of the fuction will store a 100 char string in X
a: fread(x,100,....)
b. fread(100,x,.....)
c.gets(x)
d.read(x)

27. which of the following data type will occupy the same memory irrespective of the compiler.
a.int
b.double
c.char
d.float

TI:

TEXAS INSTRUMENTS: TECHNICAL TEST

Date: 20th December 2003

1. For a CMOS inverter, the transition slope of Vout vs Vin DC characteristics can be increased (steeper transition) by:

a. Increasing W/L of PMOS transistor
b. Increasing W/L of NMOS transistor
c. Increasing W/L of both transistors by the same factor
d. Decreasing W/L of both transistor by the same factor

Ans: c

2. Minimum number of 2-input NAND gates that will be required to implement the function: Y = AB + CD + EF is

a. 4
b. 5
c. 6
d. 7

ans: c

3. Consider a two-level memory hierarchy system M1 & M2. M1 is accessed first and on miss M2 is accessed. The access of M1 is 2 nanoseconds and the miss penalty (the time to get the data from M2 in case of a miss) is 100 nanoseconds. The probability that a valid data is found in M1 is 0.97. The average memory access time is:

a. 4.94 nanoseconds
b. 3.06 nanoseconds

c. 5.00 nanoseconds
d. 5.06 nanoseconds

ans: a

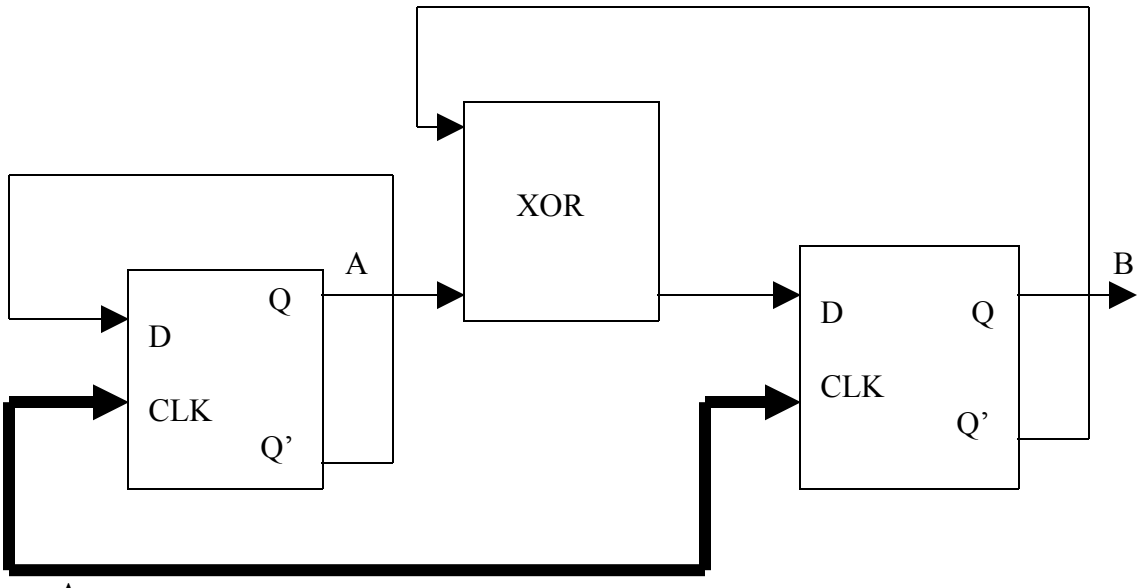
4. Interrupt latency is the time elapsed between:
   a. Occurrence of an interrupt and its detection by the CPU
   b. Assertion of an interrupt and the start of the associated ISR
   c. Assertion of an interrupt and the completion of the associated ISR
   d. Start and completion of associated ISR

Ans: d (not confirmed)

5. Which of the following is true for the function (A.B + A'.C + B.C)
   a. This function can glitch and can be further reduced
   b. This function can neither glitch nor can be further reduced
   c. This function can glitch and cannot be further reduced
   d. This function cannot glitch but can be further reduced

Ans: c **This can be reduced further using K-map, don't know abt glich, but it should not glitch**

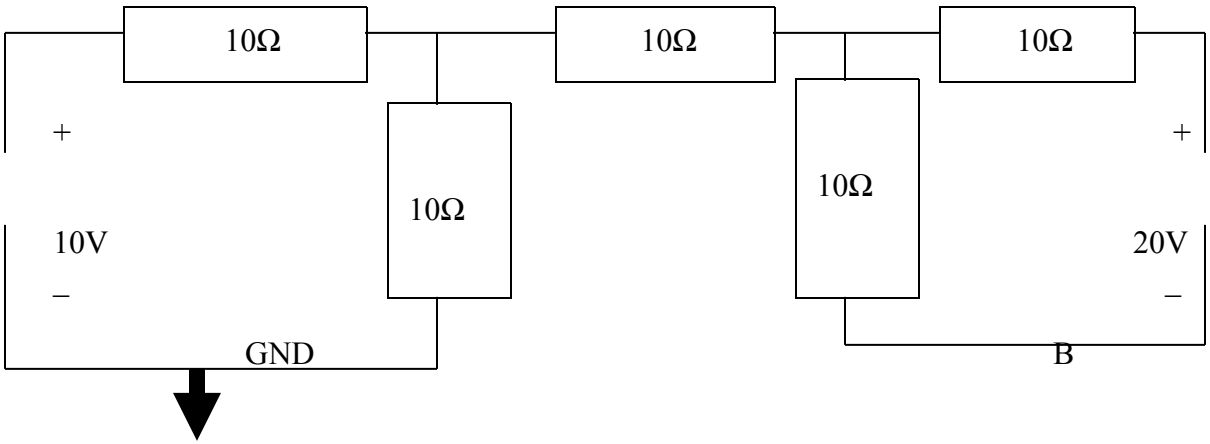
6. For the two flip-flop configuration below, what is the relationship of the output at B to the clock frequency?
   a. Output frequency is 1/4th the clock frequency, with 50% duty cycle
   b. Output frequency is 1/3rd the clock frequency, with 50% duty cycle
   c. Output frequency is 1/4th the clock frequency, with 25% duty cycle
   d. Output frequency is equal to the clock frequency



Ans: a

7. The voltage on Node B is:
   a. 0
   b. 10
   c. –10

d. –5



Ans: d

8. A CPU supports 250 instructions. Each instruction op-code has these fields:

- The instruction type (one among 250)
- A conditional register specification
- 3 register operands
- Addressing mode specification for both source operands

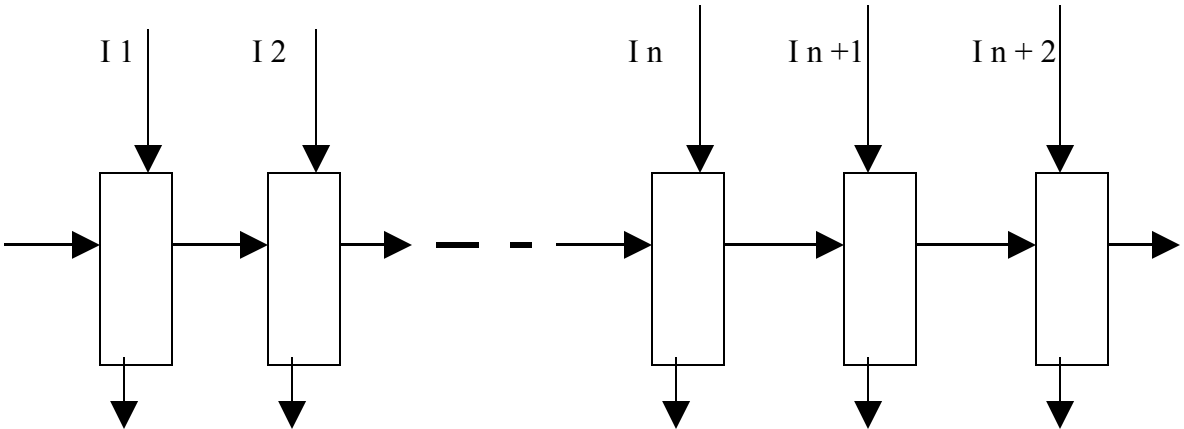
The CPU has 16 registers and supports 5 addressing modes. What is the instruction op-code length in bits?

a. 32
b. 24
c. 30
d. 36

ans: don't know

9. In the iterative network shown, the output Yn of any stage N is 1 if the total number of 1s at the inputs starting from the first stage to the Nth stage is odd. (Each identical box in the iterative network has two inputs and two outputs). The optimal logic structure for the box consists of:

a. One AND gate and one NOR gate
b. One NOR gate and one NAND gate
c. Two XNOR gates
d. One XOR gate



0

Y1 Y2 Yn Yn+1 Yn+2

Ans: d

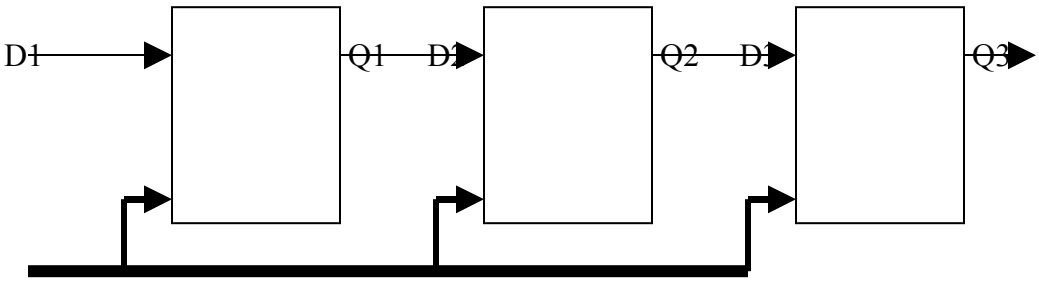
10. Consider a circuit with N logic nets. If each net can be stuck-at either values 0 and 1, in how many ways can the circuit be faulty such that only one net in it can be faulty, and such that up-to all nets in it can be faulty?

a. 2 and 2N
b. N and 2^N
c. 2N and 3^N-1
d. 2N and 3N

ans: 2N and 2^N (no match) see it .

sorry , no idea abt this

11. In the circuit shown, all the flip-flops are identical. If the set-up time is 2 ns, clock->Q delay is 3 ns and hold time is 1 ns, what is the maximum frequency of operation for the circuit?



a. 200 MHz
b. 333 MHz
c. 250 MHz
d. None of the above

Ans: a

12. Which of the following statements is/are true?

I. Combinational circuits may have feedback, sequential circuits do not.
II. Combinational circuits have a 'memory-less' property, sequential circuits do not.
III. Both combinational and sequential circuits must be controlled by an external clock.

a. I only
b. II and III only
c. I and II only
d. II only

Ans: d

13. Consider an alternate binary number representation scheme, wherein the number of ones M, in a word of N bits, is always the same. This scheme is called the M-out-of-N coding scheme. If M=N/2, and N=8, what is the efficiency of this coding scheme as against the regular binary number representation scheme? (As a hint, consider that the number of unique words represent able in the latter representation with N bits is 2^N. Hence the efficiency is 100%)

a. Close to 30%
b. Close to 50%
c. Close to 70%
d. Close to 100%

Ans: a

14. A CPU supports 4 interrupts- I1, I2, I3 and I4. It supports priority of interrupts. Nested interrupts are allowed if later interrupt is higher priority than previous one. During a certain period of time, we observe the following sequence of entry into and exit from the interrupt service routine:
I1-start---I2-start---I2-end---I4-start---I3-start---I3-end---I4-end---I1-end
From this sequence, what can we infer about the interrupt routines?

a. I3 > I4 > I2 > I1
b. I4 > I3 > I2 > I1
c. I2 > I1; I3 > I4 > I1
d. I2 > I1, I3 > I4 > I2 > I1

Ans: c

15. I decide to build myself a small electric kettle to boil my cup of tea. I need 200 ml of water for my cup of tea. Assuming that typical tap water temperature is 25 C and I want the water boiling in exactly one minute, then what is the wattage required for the heating element?
[Assume: Boiling point of water is 100 C, 1 Calorie (heat required to change 1 gm of water by 1 C)= 4 joules, 1 ml of water weighs 1 gm.]

a. Data given is insufficient
b. 800 W
c. 300 W
d. 1000 W
e. 250 W

ans: d

16. The athletics team from REC Trichy is traveling by train. The train slows down, (but does not halt) at a small wayside station that has a 100 mts long platform. The sprinter (who can run 100 mts in 10 sec) decides to jump down and get a newspaper and some idlis. He jumps out just as his compartment enters the platform and spends 5 secs buying his newspaper that is at the point where he jumped out. He then sprints along the platform to buy idlis that is another 50 mts. He spends another 5 secs buying the idlis. He is now just 50 mts from the other end of the platform where the train is moving out. He begins running in the direction of the train and the only other open door in his train is located 50 mts behind the door from where he jumped. At what(uniform) speed should the train be traveled if he just misses jumping into the open door at the very edge of the platform? Make the following assumptions

- He always runs at his peak speed uniformly
- The train travels at uniform speed
- He does not wait (other than for the idlis & newspaper) or run baclwards

a. Data given is insufficient
b. 4 m/s
c. 5 m/s
d. 7.5 m/s
e. 10 m/s

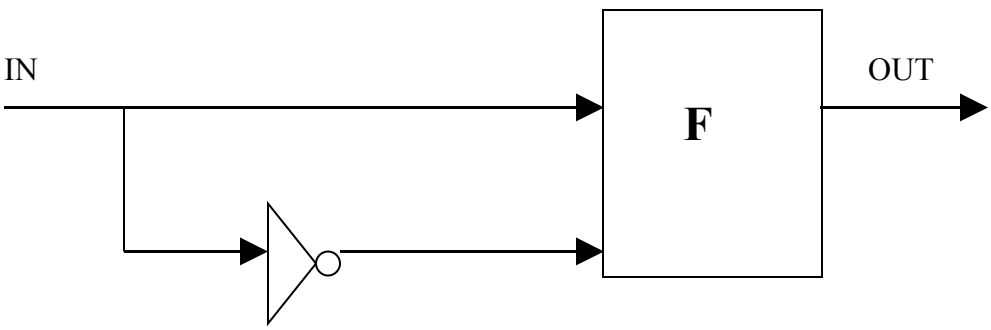
ans: c

17. State which of the following gate combinations does not form a universal logic set:

a. 2-input AND + 2-input OR
b. 2-to-1 multiplexer
c. 2-input XOR + inverter
d. 3-input NAND

ans: a

18. For the circuit shown below, what should the function F be, so that it produces an output of the same frequency (function F1), and an output of double the frequency (function F2).



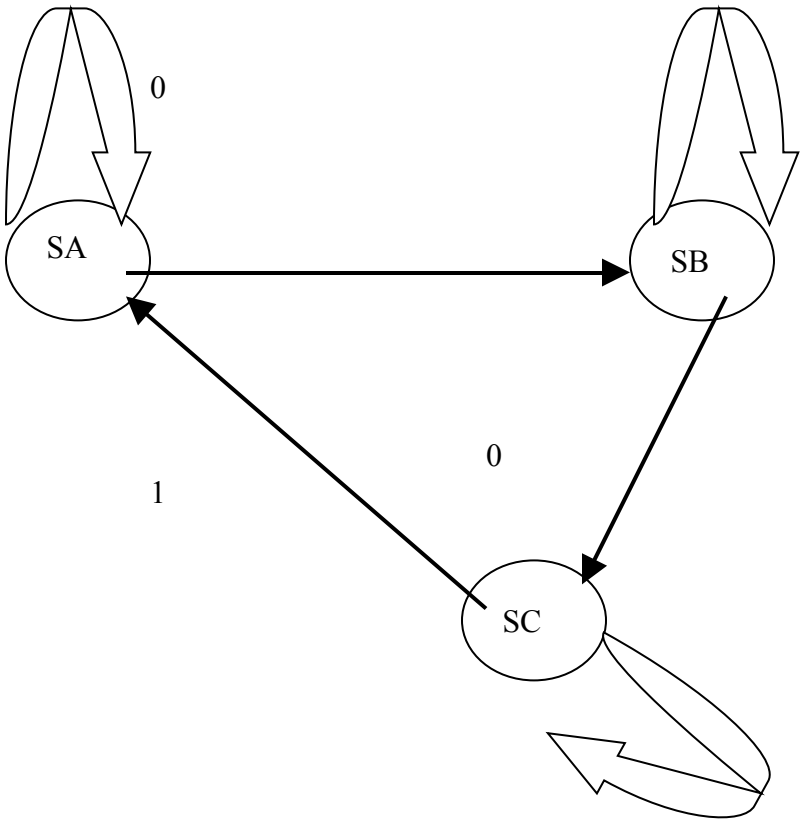
INVERTER

a. F1= NOR gate and F2= OR gate
b. F1=NAND gate and F2= AND gate
c. F1=AND gate and F2=XOR gate
d. None of the above

Ans: c

19. The FSM (finite state machine) below starts in state Sa, which is the reset state, and detects a particular sequence of inputs leading it to state Sc. FSMs have a few characteristics. An autonomous FSM has no inputs. For a Moore FSM, the output depends on the present state alone. For a Mealy FSM, the output depends on the present state as well as the inputs. Which of the statements best describes the FSM below?

a. It has two states and is autonomous
b. The information available is insufficient
c. It is a Mealy machine with three states
d. It is a Moor machine with three states

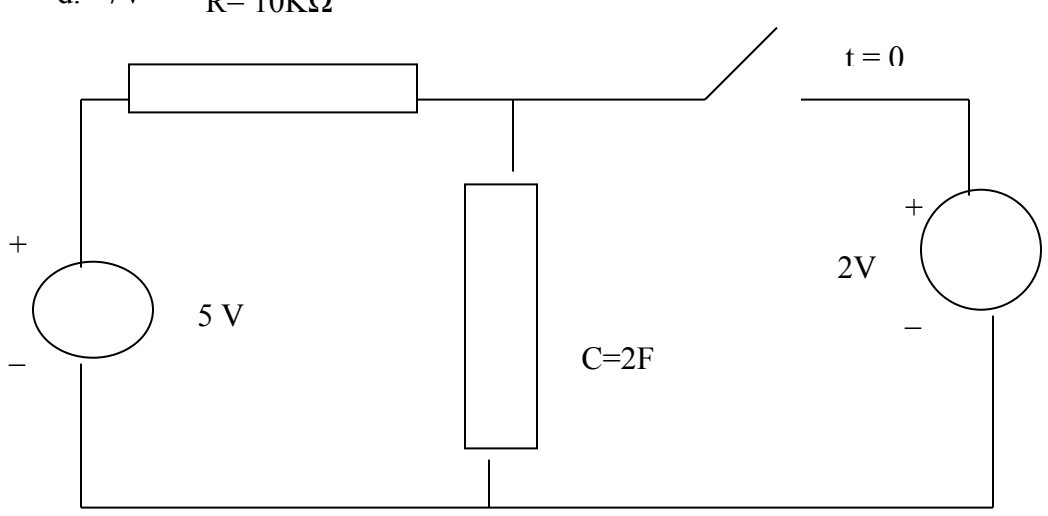


0

Ans :d

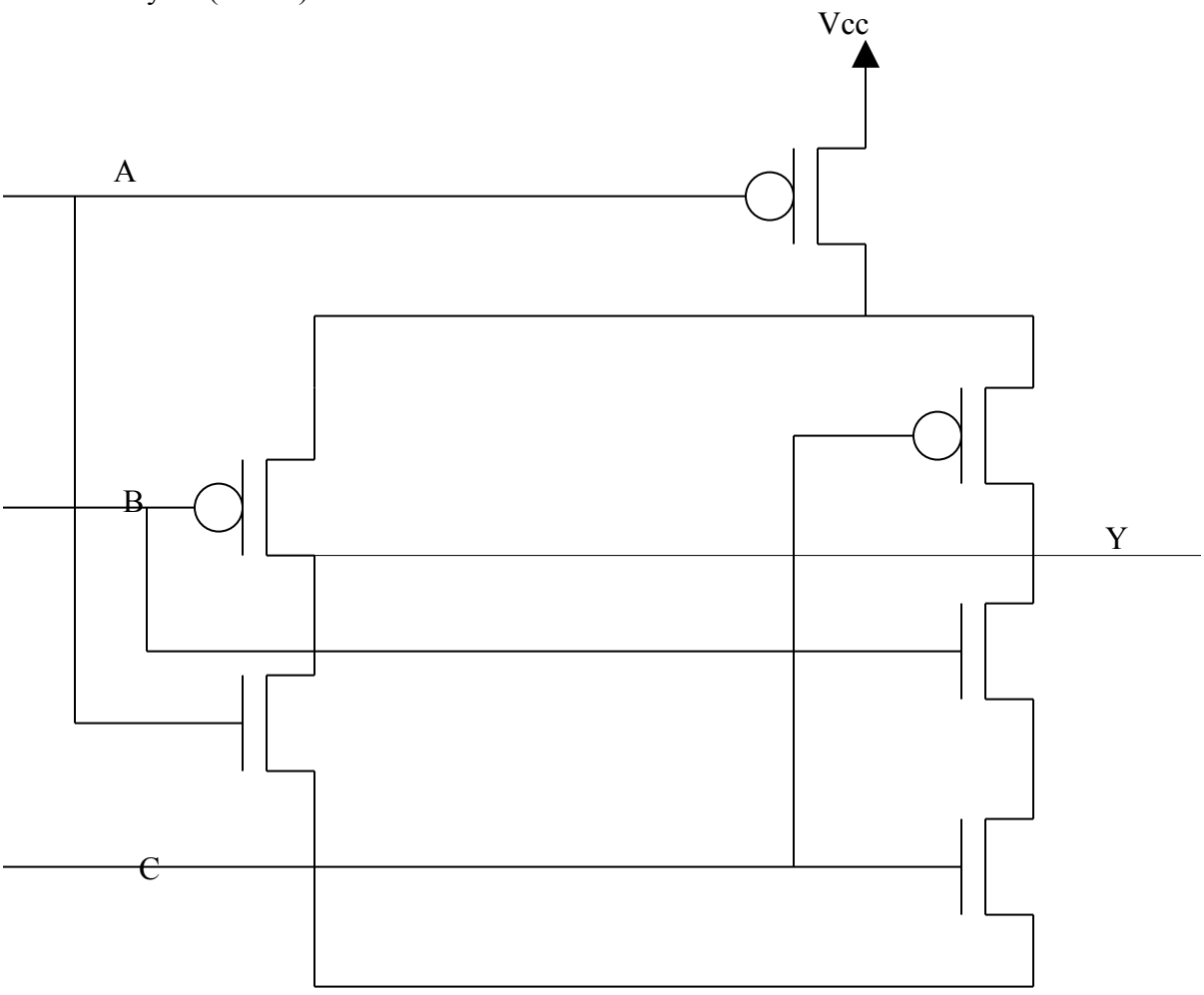
20. In the circuit given below, the switch is opened at time t=0. Voltage across the capacitor at t=infinity is:

a. 2V
b. 3V
c. 5V
d. 7V



Ans: c

21. What is the functionality represented by the following circuit?
    a. y= ! (b+ac)
    b. y= ! (a+bc)
    c. y= ! (a(b+c))
    d. y= ! (a+b+c)



Ans: b

22. The value (0xdeadbeef) needs to stored at address 0x400. Which of the below ways will the memory look like in a big endian machine:

	0x403	0x402	0x401	0x400
a.	be	ef	de	ad
b.	ef	be	ad	de
c.	fe	eb	da	ed
d.	ed	da	eb	fe

ans: don't know

ans should be (b), just check with some CS guy, little endian is Intel type, Big-endian is perhaps Motorola type

23. In a given CPU-memory sub-system, all accesses to the memory take two cycles. Accesses to memories in two consecutive cycles can therefore result in incorrect data transfer. Which of the following access mechanisms guarantees correct data transfer?

a. A read operation followed by a write operation in the next cycle.
b. A write operation followed by a read operation in the next cycle.
c. A NOP between every successive reads & writes
d. None of the above

Ans: c(not confirm)

I'm also not sure.

24. An architecture saves 4 control registers automatically on function entry (and restores them on function return). Save of each registers costs 1 cycle (so does restore). How many cycles are spent in these tasks (save and restore) while running the following un-optimized code with n=5:

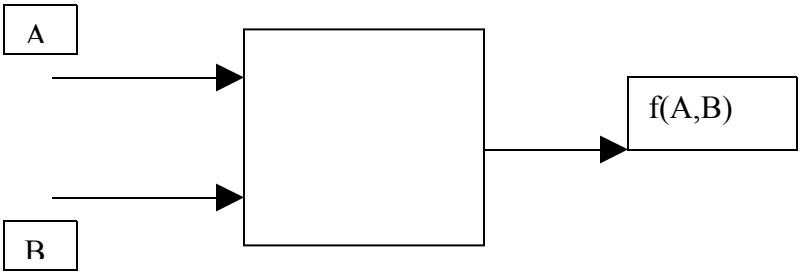
```
Void fib(int n)
{
        if((n==0) || (n==1)) return 1;
        return(fib(n-1) + fib(n-2));
}
```

a. 120
b. 80
c. 125
d. 128

ans: a

25. The maximum number of unique Boolean functions F(A,B), realizable for a two input (A,B) and single output (Z) circuit is:

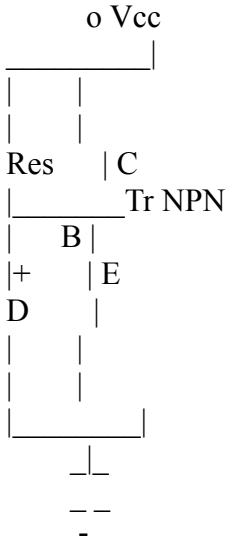
a. 2
b. 6
c. 8
d. None of the above



Ans: 2*(2*2)=16 ie d

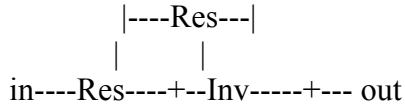
paper of TI 1999
Hard ware part only. There was one part of reasoning and there was separate paper for software persons.

1.



Find the current I delivered by the battery.

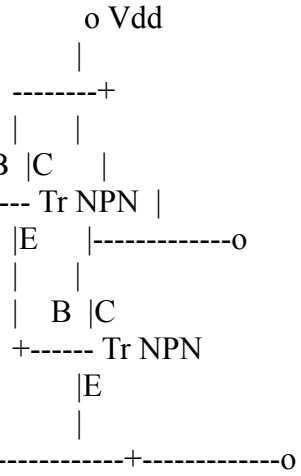
2.



What is the given circuit
a) Latch b)Amplifier c)Schmitt trigger. d)

3. The total no of flip flop required for N stage sequential circuit
N N-1 N
a)2 b)2 c) Log N d) 2 -1

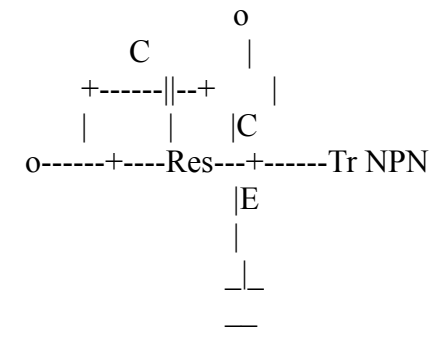
4.



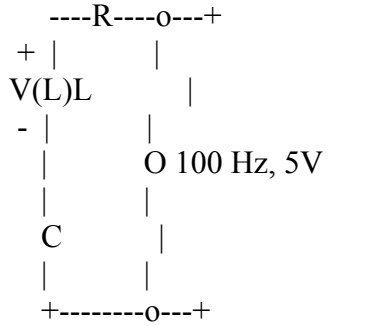
the gain of the circuit is
a) beta square b)beta + 1 c) (beta+1) ka square d)

5. If the o/p and i/p are related by y=k(x square) and i/p is a sum of 2 waveforms then the modulation scheme is
a) FM b)AM c)PM and d)None
Ans. B

6.Function of C in the circuit below is
a) Improve switching b)dc coupling c) ac coupling d) None

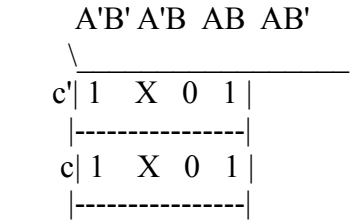


7.



if the ckt is at resonance and V(L)= (constant) V (given)
the value of V(R) and V(C) is
a)100V,5V b)-100V,5V c)5V,5V
(Use V(L)=5 /_100 and V(C)=5/_-100, V(R)=5V

8. Minimize the K-map



a) A'B' b) A'+B' c)B' d)A'+B'+C'

9. IF the rate of removal of elements in a queue containing N elements is proportional to the no of elements already existing in the queue at that instant then the no. of elements----
a)decrease linearly b)Exponetialy decrease b) Logarithmcally

10. One question on CMOS ckt.
11. Two question on OP-AMP.

THIS IS TI 1999 jadavpur for ECE students.for cs another paper is ^M
>given^M
>^M
>1.two transistors are connected Vbe is 0.7volts .this is simple ckt.one ^M
>transistor is diode equivalent. & asked the o/p across the 2 nd transistor.^M
>2.simple k map ans is Bbar.^M
>3.^M
>^M
> Emitter^M
>---R-------transistorbase| --^M
> | ---^M
> collector^M
> in above capacitor is connected parallel with resistance ^M
>r.capacitor is not shown^M
> in fig.capacitor is used for in this ckt:^M
>^M
>^M
> ans:a.speedupb.active bypass c.decoupling^M
> 4.^M
>^M
> -----R------I----------o/p^M
> |___R____ |^M
> in above r is resistence.I is cmos inverter.^M
> then ckt is used for:^M
>^M
>^M
> a.schmitt trigger b.latch c.inverter ^M

>d.amplifier^M
>^M
>^M
> 5.simple amplifier ckt openloop gain of amplifier is 4.V in ^M
>=1v.asked for V x?^M
> amplifdier + is connected to base. - is connected to i/p in between ^M
>5k is connected.^M
> from o/p feedback connected to - of amplifier with 15k.this is ckt.^M
>^M
>^M
> 6.resistence inductot cap are serially connected to ac voltage 5 ^M
>volts.voltage across^M
> inductor is given.R I C values are given & asked for^M
> voltages across resistence & capacitor.^M
> 7.^M
> ___ R_____^M
> | |^M
> ---R------OPAMP ----------^M

> |---^M
> R1 R1 is for wjhat i mean what is the purpose of R1.^M
> |^M
>^M
> ground^M
>^M
>^M
> 8.asked for Vo at the o/p.it is like simple cmos realization that is n ^M
>block is above^M
> & p block is below.Vdd is 3 volts at supply.V threshold 5 volts.^M
> 9.2 d ffs are connected in asyncro manner .clock 10 MEGAHZ.gate delay ^M
>is 1 nanosec.^M
> A B are the two given D FFs.asked for AB output is:^M
>^M
>^M
> a.updown^M
> b.up c. updown glitching like that (take care abt glitching word)^M
>^M
> 10.^M
>^M
>^M
> ----------------| subtractor|---------o/p^M
> |___HPF____|^M

>^M
> the ckt is LPF ,HPF or APF ?^M
>^M

> 11.in a queue at the no of elements removed is proportional to no of ^M
>elements in^M
> the queue.then no of elements in the queue:^M
> a.increases decreases exp or linearly(so these are the 4 options given ^M
>choose 1 option)^M
> 12.with 2 i/p AND gates u have to form a 8 i/p AND gate.which is the ^M
>fastest in the^M
> following implementations.^M
> ans we think ((AB)(CD))((EF)(GH))^M
> 13.with howmany 2:1 MUX u can for 8:1 MUX.answer is 7.^M
> 14. there are n states then ffs used are log n.^M
> 15.cube each side has r units resistence then the resistence across ^M
>diagonal of cube.^M
> 16.op amp connections asked for o/p^M
> the answer is (1+1/n)(v2-v1).check it out.practise this type of model.^M
> 17.^M
> _____ supply^M
> ---|__ ___|^M
> Ii >_____ |___ Tranistot^M
> > _____Vo^M
> > _____Vo^M
> |^M
> |^M
> R |^M
> | | Io^M
> ground.^M
>^M
>^M
>^M
>^M
> asked for Io/Ii=? transistor gain is beta.^M
>^M
>^M
> a.(1+beta)square b.1+beta c. beta^M
>^M
>^M
> 18.y=kxsquare. this is transfer function of a block with i/p x & o/p ^M
>y.if i/p is^M
> sum of a & b then o/p is :--^M
>^M
> a. AM b.FM c. PM^M
> 19.^M
> ------MULTIPLIER--- |^M

> | |^M
> _____R__|__OPAMP_____Vo^M

> ---^M
> |^M
> ground.^M
> v in = -Ez then o/p Vo =?^M
> answer is squareroot of -Ez.multiplier i/ps are a & b then ^M
>its o/p^M
> is a.b;^M

Here is Texas paper for you.
in this paper there was 20 questions as follows in 60 minutes .
second part consists of 36 que. in 30 minutes all questions are
diagramatical.(figurs)..

1. if a 5-stage pipe-line is flushed and then we have to execute 5 and 12
instructions respectively then no. of cycles will be
a. 5 and 12
b. 6 and 13
c. 9 and 16
d.none

2. k-map

ab
----
c 1 x 0 0
  1 x 0 x

solve it

a. A.B
B. ~A
C. ~B
D. A+B

3.CHAR A[10][15] AND INT B[10][15] IS DEFINED
WHAT'S THE ADDRESS OF A[3][4] AND B[3][4]
IF ADDRESS OD A IS OX1000 AND B IS 0X2000

A. 0X1030 AND 0X20C3
B. OX1031 AND OX20C4
AND SOME OTHERS..

4. int f(int *a)
{
int b=5;

```
a=&b;
}

main()
{
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}
```

what's the output .

1.10,5
2,10,10
c.5,5
d. none

```
5. main()
{
int i;
fork();
fork();
fork();
printf("----");
}
```

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

```
6.
void f(int i)
{
int j;
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
```

what's the purpose of the program

a. its output is hex representation of i
b. bcd
c. binary
d. decimal

7.#define f(a,b) a+b
#define g(a,b) a*b

main()
{

int m;
m=2*f(3,g(4,5));
printf("\n m is %d",m);
}

what's the value of m
a.70
b.50
c.26
d. 69

8.
main()
{
char a[10];
strcpy(a,"\0");
if (a==NULL)
printf("\a is null");
else
printf("\n a is not null");}

what happens with it .
a. compile time error.
b. run-time error.
c. a is null
d. a is not null.

9. char a[5]="hello"

a. in array we can't do the operation .
b. size of a is too large
c. size of a is too small
d. nothing wrong with it .

10. local variables can be store by compiler
a. in register or heap
b. in register or stack
c .in stack or heap .
d. global memory.

11. average and worst time complexity in a sorted binary tree is

12. a tree is given and ask to find its meaning (parse-tree)
(_expression tree)
ans. ((a+b)-(c*d)) (not confirmed)
13. convert 40.xxxx into binary .

14. global variable conflicts due to multiple file occurance
is resolved during
a. compile-time
b. run-time
c. link-time
d. load-time

15.
two program is given of factorial.
one with recursion and one without recursion .
question was which program won't run for very big no. input because
of stack overfow .
a. i only (ans.)
b. ii only
c. i& ii both .
c. none

16.
```
struct a
{
int a;
char b;
int c;
}

union b
{
char a;
int b;
int c;
};
```
which is correct .
a. size of a is always diff. form size of b.(ans.)

b. size of a is always same form size of b.
c. we can't say anything because of not-homogeneous (not in ordered)
d. size of a can be same if ...

TECHNICAL TEST:
---
1)3 flipflops are connected so that after 0 to 5 count occured
next number is zero. So what is the counter?
Ans: mod 6 counter

2)simplication of some boolean _expression which is simple.
Boolean _Expression is A+A'B.
Ans:A+B

3)Given inorder sequence and preorder sequence and asked to
find out postorder sequence.

4)Some question on value of a static variable.

5) Given an interger in binary form,find the number of ones in
that number without counting each bit.(This questin is not
multiple choice question. This question carries more
marks. So please take care for this question.)

6) 1-way set associative memory is called-----
a)direct b)something c)1-way set associative 4)something
Ans: c

7)Fastest IPC mechanism is

a)shared memory b)pipes c)named pipes d)semaphores
Ans:c

8)Some page references are given. You are asked to implement
it with Least Frequently Used algorithm.

9)Some diagram is given.
Iam describinmg the diagram. A 2*1 MUX is given. The inputs
are A,B. Output is C. C and A are tied together. What is
the diagram.?

Ans:Latch.

****

This paper is for Electrical & Electronics students. There is

separate test for computer Science Students. There are 20 questions.

1)Some circuit is given. Iam describing the circuit.
A resistor R & a capacitor C are connected in parallel.
To this circuit another circuit which is having a capacitor of capacity 2C & an impedence Z, is connected in series.
You are asked to find out the value of Z? Note that 2C & Z are connected in series.
a)Z=2C
b)Z=2L
c)Z=L/2
d)Z=2R

2)Some circuit which consist of only resistors R is given.
This is a repetative circuit. U have to find the effctive resistance of the entire circuit.
A)Rin=R
B)Rin=(5+sqrt(3))/7
C)Rin=(19+sqrt(3))/8
D)None.

3)Two wave forms are given. You are asked to write the cirsuit to get B(second wave form) from A(first wave form).

4)#define SUM(a,b) a+b

```
main()
{
a=2;
b=3;
x=SUM(a,b)*2;
printf("x=%d\n",x);
}
```
Ans:8.

5)
```
number(int i)
{
number++;
printf("%d\n",number);
}

main()
{
static int i=0;
number(i);
```

}
Ans: I don't know.

6)Some circuit is given. I can't describe the circuit. There are
3 resistors,3 capacitors & one inverter.. The question is
What is the value of the frequency such that the circuit oscillates.
A)f=RC
B)f=sqrt(3)/(Pi*R*C)
C)f=1/(Pi*R*C)
D)something
Ans:I don't know the answer.

7)Question on flipflop. So gothrough all flipflops.

8)There are 5 questions on Nmos & Pmos circuits.

****
This Paper is for Computer Science Students. THis paper is
very easy. You can definitely do it in one hour.
****

(1) The fastest memory is
(i) DRAM, (ii) ROM, (iii) SRAM, (iv) Main memory
Ans : SRAM

(2) Programing exceptions are
(i) Asynchronous, (ii) Synchronous, (iii) None
Ans : Asynchronous

(3) DSP which architecture is used
(i) MIMD, (ii) SIMD, (iii) Nueman, (iv) Harvard Architecture
Ans : Harvard Architecture

(4) C prog. for searching for an element in linked list

(5) main()
{
unsigned char i;
int sum;

for(i=0; i<300; i++)
sum+ = i;
printf("\nSum = %d\n", sum);
}

Ans : infinite loop

(6) void fn(int *p)
{
static int val = 100;
p = &val;
}

main()
{
int i=10;
printf("i=%d\n", i);
fn(&i);
printf("i=%d\n", i);
}

Ans : i=10 i=10

(7) int a[10[15];
char b[10[15];
(a) location g a[3][4], if base location g a[0][0] is ox1000
(b) location g b[3][4], if base location g b[0][0] is ox2000
int taken 32 bits and char taken 8 bits.

Ans : (a) ox10C4 (b) ox2031

(8) Implement OR gate function with 2*1 MUX

Ans : A _____
--------|2*1 MUX |
B | |--------o/p
--------| |
| -----------
|_____|C

B=C

(9) Implement 4*1 MUX with 2*1 MUXES

(10) Swapping without using a temporary variables. (2 methods)

(i) x = x+y;
y = x-y;
x = x-y;

(ii) x = x^y;

```
y = x^y;
x = x^y;
```

(11) Count no of 1's in a word without using bit by bit.
(This question carries more marks. It is not a multiple choice question.)

(12) Code 1 :

```
for(i=0; i<1000; i++)
for(j=0; j<100; j++)
x = y;
```

Code 2 :

```
for(i=0; i<100; i++)
for(j=0; j<1000; j++)
x = y;
```

Which code will execute faster
(i) Code 1 and Code 2 are of same speed,
(ii) Code 1,
(iii) Code 2,
(iv) None.

Ans : Code 2

(13)

```
main()
{
int a[10] = {1, 2, 3, ...., 10}, i, x=10, temp;

for(i=0; i temp = a[i];
a[i] = a[x-i-1];
a[x-i-1] = temp;
}
```

(i) All contents of array a are reversed
(ii) Only some portions are altered
(iii) Remains same
(iv) None

Ans : (iii)

(14) An array is stored in row major order. The memory capacity is 30 MB. And in unix system demand paging is used. Which one will give more page faults?

```
#define V_L_I 10000
int i, j, array[V_L_I][V_L_I];
```

Code 1 :
array[i][j] = 1;

Code 1 :
for(j=0; j for(i=0; i array[i][j] = 1;

Ans : Code 2

(15) In C which parameter passing technique is used?
(i) call by value,
(ii) call by reference,
(iii) both

Ans : call by value

(16) A circuit is given with 2 exclusive OR gates whose boolean
_expression will be y = '(AB) + AB
(' indicates bar)

(17) main()
{
int i = 1;
fork();
fork();
printf("\ni = %d\n", i+1);
}

Ans : 4 printfs will occur and i = 2

(18) Compute the complexity of Binary search.
Ans : O(lg n) (Answer in detail. This is not a multiple choice question.
It carries more marks.)

(19) Write _expression for the tree graph :
Ans : ((a-b) + c*d)/x

(20) # define MAX(a, b) a>b ? a:b
main()
{
int m, n;
m = 3 + MAX(2, 3);
n = 2 * MAX(3, 2);
printf("m = %d, n = %d\n", m, n)
}

Ans : m=2, n=3
paper of texas instruments.
Technical + aptitude + interview.
* Questions on c
here i am not strictly following syntax it is just to show what was asked.
1 #define sum(a,b) a+b
value of sum(2,3)*2 Ans:8

2. a=5,b=6
swap(&a,b);
This function is written to swap a and b
find value of a and b . Ans 6,6

3.
```
function()
{
static int a=0;
a=a+1;
return a;
}
main()
{
function();
function();
function();
printf a;
}
```
final value of a ? Ans : a=3. static initializes once.
4.Write two prog. to swap a & b without using temp variable.
5.
```
unsigned char i;
int sum=0;
for(i=0;i<300;i++)
sum=sum+i;
printf(sum);
```

Ans:Program will held in infinite loop b/c i can not exceed 255.

6.
five questions on MOSFETS.
four were having single mosfets.
questions were simple.as i told u in Gwalior.
just on the funda that it will conduct if Vg-Vs > Vt .

In one question output at drain was to be calculated while o/p was initially charged to 5v and to the gate 5v were applied.

In one question output at drain was to be calculated while o/p was initially charged to 5v and the gate was shorted to drain.
7.
Clear the concept of settling time , hold time and other times. 3 ques on that.
like values of various delays were given and max frequency at which the circuit can work
hint : 1/sum of all delays . In our case ans was 200 Mhz.

8. An input and output waveform was given and circuit was to be designed with the use of one
delay.
Ans : exor gate in which second input is first input with a delay.

9. A question to determine sequence of counter. Don't get puzzled it was a tough question.

10.The output and input of a inverter is connected by three RC stages in between of each stage two amplifiers with poles at imaginary axis were connected.
Hint : The poles at imaginary axis will create extra 180 phase shift thus the circuit will oscillate and calculate the frequency of operation.

11. A series of infinite connected rc circuit and overall input resistance is calculated.(question of 12 class)

```
        TECHNICAL TEST:
        ------------------------

        1)3 flipflops are connected so that after 0 to 5 count occured
          next number is zero. So what is the counter?
          Ans: mod 6 counter

   2)simplication of some boolean expression which is simple.
     Boolean Expression is A+A'B.
     Ans:A+B

   3)Given inorder sequence and preorder sequence and asked to
       find out postorder sequence.

  4)Some question on value of a static variable.

  5) Given an interger in binary form,find the number of ones in
        that number without counting each bit.(This questin is not
              multiple choice question. This question carries more
              marks. So please take care for this question.)

6) 1-way set associative memory is called-----
    a)direct b)something c)1-way set associative 4)something
    Ans: c

7)Fastest IPC mechanism is
```

a)shared memory b)pipes c)named pipes d)semaphores
Ans:c

8)Some page references are given. You are asked to implement
it with Least Frequently Used algorithm.

9)Some diagram is given.
Iam describinmg the diagram. A 2*1 MUX is given. The inputs
are A,B. Output is C. C and A are tied together. What is
the diagram.?

Ans:Latch.

****

This paper is for Electrical & Electronics students. There is
separate test for computer Science Students. There are 20
questions.

1)Some circuit is given. Iam describing the circuit.
A resistor R & a capacitor C are connected in parallel.
To this circuit another circuit which is having a capacitor
of capacity 2C & an impedence Z, is connected in series.
You are asked to find out the value of Z? Note that 2C & Z
are connected in series.
a)Z=2C
b)Z=2L
c)Z=L/2
d)Z=2R

2)Some circuit which consist of only resistors R is given.
This is a repetative circuit. U have to find the effctive
resistance of the entire circuit.
A)Rin=R
B)Rin=(5+sqrt(3))/7
C)Rin=(19+sqrt(3))/8
D)None.

3)Two wave forms are given. You are asked to write the cirsuit
to get B(second wave form) from A(first wave form).

4)
```
#define SUM(a,b)  a+b

main()
{
  a=2;
  b=3;
  x=SUM(a,b)*2;
  printf("x=%d\n",x);
  }
```
Ans:8.

5)
```
number(int i)
{
number++;
printf("%d\n",number);
```

```
}

main()
{
static int i=0;
number(i);
}
```
Ans: I don't know.

6) Some circuit is given. I can't describe the circuit. There are
3 resistors,3 capacitors & one inverter.. The question is
What is the value of the frequency such that the circuit oscillates.
A) f=RC
B) f=sqrt(3)/(Pi*R*C)
C) f=1/(Pi*R*C)
D) something
Ans:I don't know the answer.

7) Question on flipflop. So gothrough all flipflops.

8) There are 5 questions on Nmos & Pmos circuits.

****
This Paper is for Computer Science Students. THis paper is
very easy. You can definitely do it in one hour.
****

(1) The fastest memory is
(i) DRAM, (ii) ROM, (iii) SRAM, (iv) Main memory
Ans : SRAM

(2) Programing exceptions are
(i) Asynchronous, (ii) Synchronous, (iii) None
Ans : Asynchronous

(3) DSP which architecture is used
(i) MIMD, (ii) SIMD, (iii) Nueman, (iv) Harvard Architecture
Ans : Harvard Architecture

(4) C prog. for searching for an element in linked list

(5)
```
main()
{
        unsigned char   i;
        int     sum;

        for(i=0; i<300; i++)
                sum+ = i;
        printf("\nSum = %d\n", sum);
}
```

Ans : infinite loop

(6)
```
void fn(int *p)
        {
                static int val = 100;
                p = &val;
```

```
        }

main()
{
        int i=10;
        printf("i=%d\n", i);
        fn(&i);
        printf("i=%d\n", i);
}
```

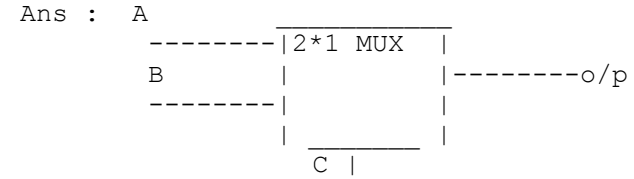
Ans : i=10 i=10

(7) int a[10[15];
        char b[10[15];
        (a) location g a[3][4], if base location g a[0][0] is ox1000
        (b) location g b[3][4], if base location g b[0][0] is ox2000
        int taken 32 bits and char taken 8 bits.

Ans : (a) ox10C4   (b) ox2031

(8) Implement OR gate function with 2*1 MUX

Ans :



B=C

(9) Implement 4*1 MUX with 2*1 MUXES

(10) Swapping without using a temporary variables. (2 methods)

(i) x = x+y;
        y = x-y;
        x = x-y;

(ii) x = x^y;
         y = x^y;
         x = x^y;

(11) Count no of 1's in a word without using bit by bit.

>^M
>   THIS IS TI 1999 jadavpur for ECE students.for cs another paper is ^M
>given^M
>^M
>1.two transistors are connected Vbe is 0.7volts .this is simple ckt.one ^M
>transistor is diode equivalent. & asked the o/p across the 2 nd transistor.^M

>2.simple k map ans is Bbar.^M
>3.^M
>^M
>   Emitter^M
>---R-------transistorbase| --^M
> | ---^M
>   collector^M
>   in above capacitor is connected parallel with resistance
^M
>r.capacitor is not shown^M
>   in fig.capacitor is used for in this ckt:^M
>^M
>^M
>   ans:a.speedupb.active bypass c.decoupling^M
>   4.^M
>^M
>   -----R------I----------o/p^M
> |___R____ |^M
>   in above r is resistence.I is cmos
inverter.^M
>   then ckt is used for:^M
>^M
>^M
>   a.schmitt trigger b.latch c.inverter ^M
>d.amplifier^M
>^M
>^M
>   5.simple amplifier ckt openloop gain of amplifier is 4.V in ^M
>=1v.asked for V x?^M
>   amplifdier + is connected to base. - is connected to i/p in
between ^M
>5k is connected.^M
>   from o/p feedback connected to - of amplifier with 15k.this is
ckt.^M
>^M
>^M
>   6.resistence inductot cap are serially connected to ac voltage 5
^M
>volts.voltage across^M
>   inductor is given.R I C values are given & asked for^M
>   voltages across resistence & capacitor.^M
>   7.^M
>   ___ R_____^M
> | |^M
>   ---R------OPAMP ----------^M

                  > |---^M
>   R1   R1 is for wjhat i mean what is the purpose of
R1.^M
> |^M
>^M
>   ground^M
>^M
>^M
>   8.asked for Vo at the o/p.it is like simple cmos realization that
is n ^M
>block is above^M

>   & p block is below.Vdd is 3 volts at supply.V threshold 5 volts.^M
>   9.2 d ffs are connected in asyncro manner .clock 10 MEGAHZ.gate
delay ^M
>is 1 nanosec.^M
>   A B are the two given D FFs.asked for AB output is:^M
>^M
>^M
>   a.updown^M
>   b.up c. updown glitching like that (take care abt glitching
word)^M
>^M
>   10.^M
>^M
>^M
>   ----------------| subtractor|---------o/p^M
> |___HPF____|^M

                        >^M
>   the ckt is LPF ,HPF or APF ?^M
>^M
>   11.in a queue at the no of elements removed is proportional to no
of ^M
>elements in^M
>   the queue.then no of elements in the queue:^M
>   a.increases decreases exp or linearly(so these are the 4 options
given ^M
>choose 1 option)^M
>   12.with 2 i/p AND gates u have to form a 8 i/p AND gate.which is
the ^M
>fastest in the^M
>   following implementations.^M
>   ans we think ((AB)(CD))((EF)(GH))^M
>   13.with howmany 2:1 MUX u can for   8:1 MUX.answer is 7.^M
>   14. there are n states then ffs used are log n.^M
>   15.cube each side has r units resistence then the resistence across
^M
>diagonal of cube.^M
>   16.op amp connections asked for o/p^M
>   the answer is (1+1/n)(v2-v1).check it out.practise this type of
model.^M
>   17.^M
>   _____________ supply^M
>   ---|__   ___|^M
> Ii   >_________ |___   Tranistot^M
>   > _______Vo^M
>   > _______Vo^M
> |^M
> |^M
>   R |^M
> | | Io^M
>   ground.^M
>^M
>^M
>^M
>^M
>   asked for Io/Ii=? transistor gain is beta.^M
>^M

>^M
>   a.(1+beta)square b.1+beta c. beta^M
>^M
>^M
>   18.y=kxsquare. this is transfer function of a block with i/p x & o/p ^M
>y.if i/p is^M
>   sum of a & b then o/p is :--^M
>^M
>   a. AM b.FM c. PM^M
>   19.^M
>   ------MULTIPLIER--- |^M

                       > | |^M
>   _____R__|__OPAMP_____________________Vo^M
>   ---^M
> |^M
>   ground.^M
>   v in = -Ez then o/p Vo =?^M
>   answer is squareroot of -Ez.multiplier i/ps are a & b then ^M
>its o/p^M
>   is a.b;^M

Here is Texas paper for you.

in this paper there was 20 questions as follows in 60 minutes .
second part consists of 36 que. in 30 minutes all questions are diagramatical.(figurs)..

1. if a 5-stage pipe-line is flushed and then we have to execute 5 and
12
instructions respectively then no. of cycles will be
a. 5 and 12
b. 6 and 13
c. 9 and 16
d.none

2. k-map

ab
---
c   1   x 0 0
       1   x 0 x

solve it

a. A.B
B. ~A
C. ~B
D. A+B

3.CHAR A[10][15] AND INT B[10][15] IS DEFINED
WHAT'S THE ADDRESS OF A[3][4] AND B[3][4]
IF ADDRESS OD A IS OX1000 AND B IS 0X2000

A. 0X1030 AND 0X20C3

B. OX1031 AND OX20C4
AND SOME OTHERS..

```
4. int f(int *a)
{
int b=5;
a=&b;
}

main()
{
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}
```

what's the output .

1.10,5
2,10,10
c.5,5
d. none

```
5. main()
{
int i;
fork();
fork();
fork();
printf("----");
}
```

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

```
6.
void f(int i)
{
int j;
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
```
what's the purpose of the program

a. its output is hex representation of i
b. bcd
c. binary
d. decimal

7.
```
#define f(a,b) a+b
#define g(a,b) a*b

main()
{

int m;
m=2*f(3,g(4,5));
printf("\n m is %d",m);
}
```

what's the value of m
a.70
b.50
c.26
d. 69

8.
```
main()
{
char a[10];
strcpy(a,"\0");
if (a==NULL)
printf("\a is null");
else
printf("\n a is not null");}
```

what happens with it .
a. compile time error.
b. run-time error.
c. a is null
d. a is not null.

9. `char a[5]="hello"`

a. in array we can't do the operation .
b. size of a is too large
c. size of a is too small
d. nothing wrong with it .

10. local variables can be store by compiler
a. in register or heap
b. in register or stack
c .in stack or heap .
d. global memory.

11. average and worst time complexity in a sorted binary tree is

12. a tree is given and ask to find its meaning (parse-tree)
(expression tree)
ans. ((a+b)-(c*d)) (not confirmed)
13. convert 40.xxxx into binary .

14. global variable conflicts due to multiple file occurance
is resolved during
a. compile-time

b. run-time
c. link-time
d. load-time

15.
two program is given of factorial.
one with recursion and one without recursion .
question was which program won't run for very big no. input
because
of stack overfow .
a. i only (ans.)
b. ii only
c. i& ii both .
c. none

16.
struct a
{
int a;
char b;
int c;
}

union b
{
char a;
int b;
int c;
};
which is correct .
a. size of a is always diff. form size of b.(ans.)
b. size of a is always same form size of b.
c. we can't say anything because of not-homogeneous (not in ordered)
d. size of a can be same if ...

Hi friends

here is the full paper of TI India IITB 20/07/2001.
WITH SOME ANSWER

the paper had 4 sections in which we have to attend
2 sections
i attempted digtal and analog

and in apti, there was 75 Qs in 60min. be prepare
for
it also, apti was very tough. Apti was more
mathematical stuff, requires a
lot of thinking and very high speed. ex. coding,
reading compreh.(RC),
calculations, relationship, travelling problem
(given some cities and
journey conditions were given) etc etc...

Best of luck
Regards

Vijay Mathur

DIGITAL

1. nand gate is
a) associative &cumulative b)cumulative but not associative
c)not cumulative but associative d)not cumultive and associative

ANS. b

2. which imp has les delay
a) (a xor b) xor (c xor d)
b) (((a xor b) xor c) xor d)
(think on the situation when input a, b, c, d are comes in ascending (i.e. a frist then b then c and then d) and desending (opposite order)

3) one inverter cmos circuit was given with A variable and enable B signal

ANS. tristate inverter with B as enable

4)a logic cell which derterminesⓄp =1) for odd no.. of 1s in the given seq is

ANS. one xor gate

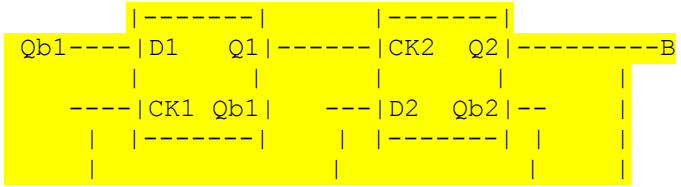
5)circuit

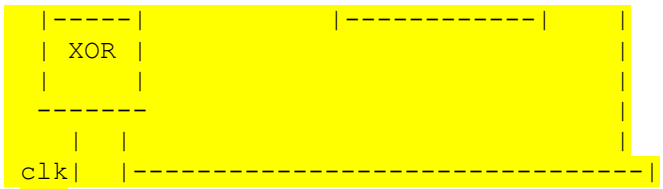
```
------
| ttl |q---+diode--inverter--res--+led---gnd
|Logic|
-------
```

led should glow when q=0 and off when q=1
the choices are
a.ckt will funct as given
b.it wont funct as given
c.q cant drive ttl inverter
d.non of these
ANS: b

6)n nets are givenin how many ways can we model each
of the stuck at fault in n nets in single(one at a time) and w....
ANS. 2n and (3^n)-1

7) circiut with 2 d ffs was given





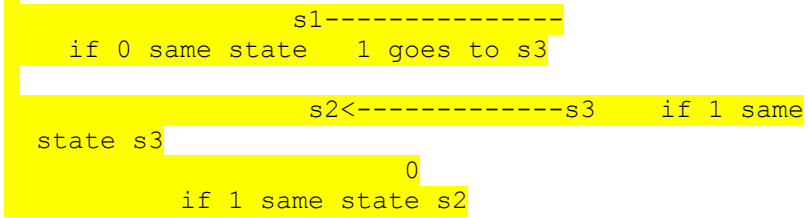
whaT is the relation between B and clk i/p?

Ans. b is 1/3 of clk i/p with 50% duty cycle.

8)3 dffs was given with common clk
setup time 3ns
hold time 1ns
clk to q delay 2ns find the maximum frequency of operation

ANS: 200MHz

9)fsm question there states given ques what is the
machne called



ans: MOORE M/C

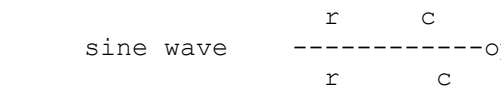
10. consider a adder and multiplier
question ios some what like can both implemented in
same fsm or cant

ANS: Yes they can implemented in same machine.

these are the 10 ques asked in digitl section and we are in a
situation to attend 2 section i attented analog part
and i am giving the ruogh idea of analog section

****
Analog part 10questions

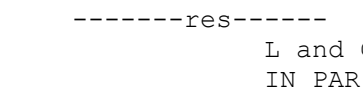
1. input is sine wave
what is the op in the middle of serially connected 2
rc parallel ckts



a. cos wave
b.sine wave with 0 phase shift
etc

ANS: b

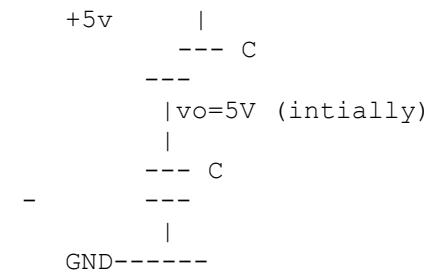
2.square wave as input given to the ckt



What is the op in the cap

ANS: SINE WAVE.

3.in 5v wave



to the circuit in which two cap are connected in series

what is the op at the vo at time T
intially it was at 5v

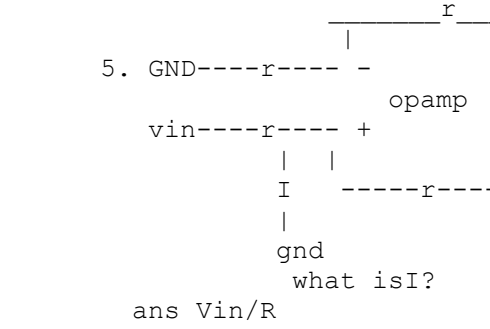
ans: may be 5V. (CHECK IT MAY BE 2.5 V)

4.darlinton pair was given with beta as ct gain of each trans

what is the overall gain

ans: (beta+1)^2

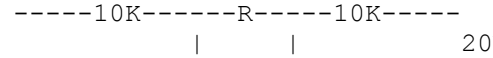
5.

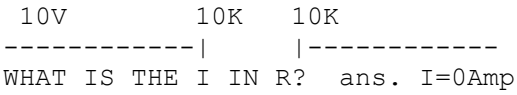


what isI?

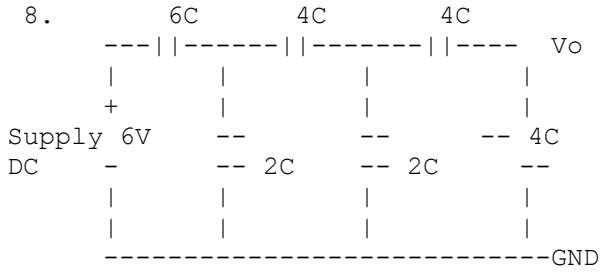
ans Vin/R

6.

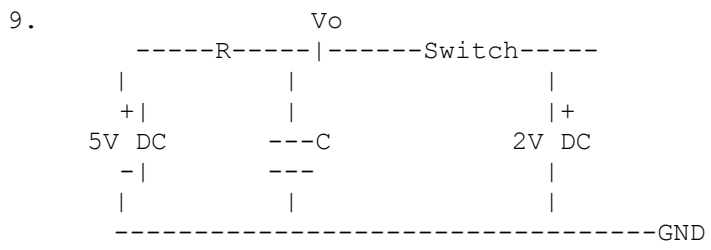




7. IF THE INPUT SGL IS 95khz and it is sampled at 120samples per sec the at what freq wil the fft opt fundemental freq will come
totally there were 10 questions in analog ssection also
the numbering is not right



find out Vo=? (caps vaule may be changed).
ANS was 0.75V but here i think cap. values changed, so calculate urself
for currect answer.



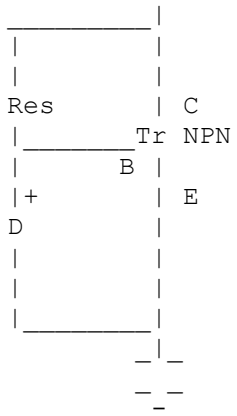
Switch is open at t=0, what is the value of Vo at t=infinity. Ans. 5V

10. one Qs on current mirror. there are n current mirrors are connected in series, you have to find out
the condition for whitch current mirrors will be in linear region.
(a)Vt
(b)Vt+ deta V
(c)n*Vt+ (n-1)* delta V
(d)n(Vt+delta V)

ANS: c

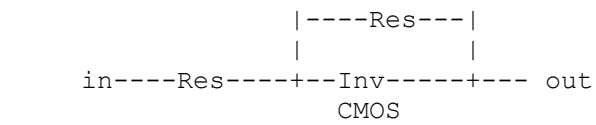
paper of TI 1999
Hard ware part only. There was one part of reasoning and there was separate paper for software persons.

1. o Vcc



Find the current I delivered by the battery.

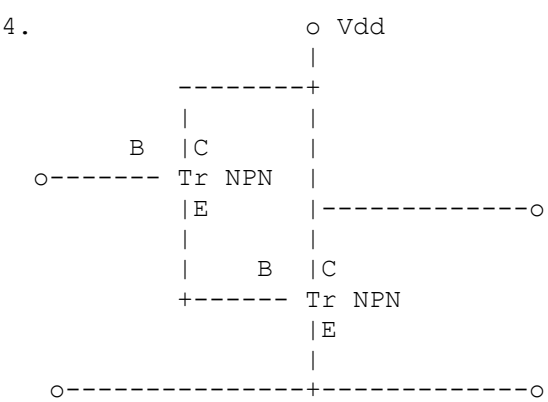
2.



What is the given circuit
a) Latch b)Amplifier c)Schmitt trigger. d)

3. The total no of flip flop required for N stage sequential circuit
N N-1 N
a)2 b)2 c) Log N d) 2 -1

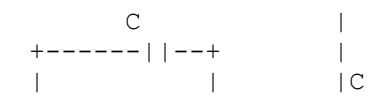
4.

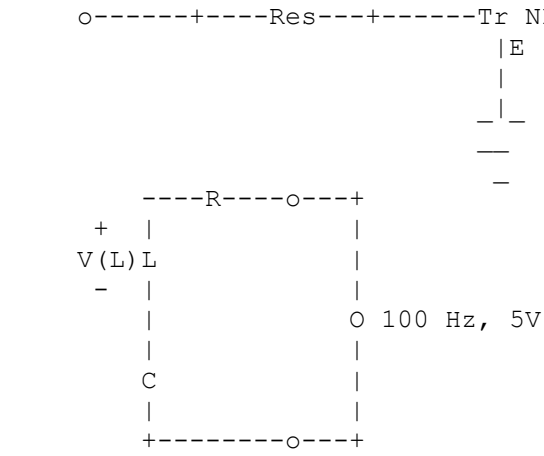


the gain of the circuit is
a) beta square b)beta + 1 c) (beta+1) ka square d)

5. If the o/p and i/p are related by y=k(x square) and i/p is a sum of 2 waveforms
then the modulation scheme is
a) FM b)AM c)PM and d)None
Ans. B

6.Function of C in the circuit below is
a) Improve switching b)dc coupling c) ac coupling d) None





7. if the ckt is at resonance and V(L)= (constant) V (given)
the value of V(R) and V(C) is
a)100V,5V b)-100V,5V c)5V,5V
(Use V(L)=5 /_100 and V(C)=5/_-100, V(R)=5V

8. Minimize the K-map

	A'B'	A'B	AB	AB'
c'	1	X	0	1
c	1	X	0	1

a) A'B' b) A'+B' c)B' d)A'+B'+C'

9. IF the rate of removal of elements in a queue containing N elements is
proportional to the no of elements already existing in the queue at that
instant then the no. of elements----
a)decrease linearly b)Exponetialy decrease b) Logarithmcally

10. One question on CMOS ckt.
11. Two question on OP-AMP.

THIS IS TI 1999 jadavpur for ECE students.for cs another paper is ^M
>given^M
>^M
>1.two transistors are connected Vbe is 0.7volts .this is simple ckt.one ^M
>transistor is diode equivalent. & asked the o/p across the 2 nd transistor.^M
>2.simple k map ans is Bbar.^M
>3.^M
>^M
> Emitter^M
>---R-------transistorbase| --^M
> | ---^M
> collector^M
> in above capacitor is connected parallel with resistance ^M
>r.capacitor is not shown^M
> in fig.capacitor is used for in this ckt:^M

>^M
>^M
> ans:a.speedupb.active bypass c.decoupling^M
> 4.^M
>^M
> -----R------I----------o/p^M
> |___R____ |^M
> in above r is resistence.I is cmos inverter.^M
> then ckt is used for:^M
>^M
>^M
> a.schmitt trigger b.latch c.inverter ^M
>d.amplifier^M
>^M
>^M
> 5.simple amplifier ckt openloop gain of amplifier is 4.V in ^M
>=1v.asked for V x?^M
> amplifdier + is connected to base. - is connected to i/p in between ^M
>5k is connected.^M
> from o/p feedback connected to - of amplifier with 15k.this is ckt.^M
>^M
>^M
> 6.resistence inductot cap are serially connected to ac voltage 5 ^M
>volts.voltage across^M
> inductor is given.R I C values are given & asked for^M
> voltages across resistence & capacitor.^M
> 7.^M
> ___ R_____^M
> | |^M
> ---R------OPAMP ----------^M

> |---^M
> R1 R1 is for wjhat i mean what is the purpose of R1.^M
> |^M
>^M
> ground^M
>^M
>^M
> 8.asked for Vo at the o/p.it is like simple cmos realization that is
n ^M
>block is above^M
> & p block is below.Vdd is 3 volts at supply.V threshold 5 volts.^M
> 9.2 d ffs are connected in asyncro manner .clock 10 MEGAHZ.gate delay
^M
>is 1 nanosec.^M
> A B are the two given D FFs.asked for AB output is:^M
>^M
>^M
> a.updown^M
> b.up c. updown glitching like that (take care abt glitching word)^M
>^M
> 10.^M
>^M
>^M
> ----------------| subtractor|---------o/p^M
> |___HPF____|^M

>^M
> the ckt is LPF ,HPF or APF ?^M
>^M
> 11.in a queue at the no of elements removed is proportional to no of
^M
>elements in^M
> the queue.then no of elements in the queue:^M
> a.increases decreases exp or linearly(so these are the 4 options
given ^M
>choose 1 option)^M
> 12.with 2 i/p AND gates u have to form a 8 i/p AND gate.which is the
^M
>fastest in the^M
> following implementations.^M
> ans we think ((AB)(CD))((EF)(GH))^M
> 13.with howmany 2:1 MUX u can for 8:1 MUX.answer is 7.^M
> 14. there are n states then ffs used are log n.^M
> 15.cube each side has r units resistence then the resistence across ^M
>diagonal of cube.^M
> 16.op amp connections asked for o/p^M
> the answer is (1+1/n)(v2-v1).check it out.practise this type of
model.^M
> 17.^M
> _____________ supply^M
> ---|__ ___|^M
> Ii >_________ |___ Tranistot^M
> > ________Vo^M
> > ________Vo^M
> |^M
> |^M
> R |^M
> | | Io^M
> ground.^M
>^M
>^M
>^M
>^M
> asked for Io/Ii=? transistor gain is beta.^M
>^M
>^M
> a.(1+beta)square b.1+beta c. beta^M
>^M
>^M
> 18.y=kxsquare. this is transfer function of a block with i/p x & o/p
^M
>y.if i/p is^M
> sum of a & b then o/p is :--^M
>^M
> a. AM b.FM c. PM^M
> 19.^M
> ------MULTIPLIER--- |^M

> | |^M
> _____R__|__OPAMP_____________________Vo^M
> ---^M
> |^M
> ground.^M

> v in = -Ez then o/p Vo =?^M
> answer is squareroot of -Ez.multiplier i/ps are a & b then ^M
>its o/p^M
> is a.b;^M

Here is Texas paper for you.
in this paper there was 20 questions as follows in 60 minutes .
second part consists of 36 que. in 30 minutes all questions are
diagramatical.(figurs)..

1. if a 5-stage pipe-line is flushed and then we have to execute 5 and
12
instructions respectively then no. of cycles will be
a. 5 and 12
b. 6 and 13
c. 9 and 16
d.none

2. k-map

ab
----
c 1 x 0 0
1 x 0 x

solve it

a. A.B
B. ~A
C. ~B
D. A+B

3.CHAR A[10][15] AND INT B[10][15] IS DEFINED
WHAT'S THE ADDRESS OF A[3][4] AND B[3][4]
IF ADDRESS OD A IS OX1000 AND B IS 0X2000

A. 0X1030 AND 0X20C3
B. OX1031 AND OX20C4
AND SOME OTHERS..

```
4. int f(int *a)
{
int b=5;
a=&b;
}

main()
{
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}
```

what's the output .

1.10,5

2,10,10
c.5,5
d. none

5. main()
{
int i;
fork();
fork();
fork();
printf("----");
}

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

6.
void f(int i)
{
int j;
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
what's the purpose of the program

a. its output is hex representation of i
b. bcd
c. binary
d. decimal

7.#define f(a,b) a+b
#define g(a,b) a*b

main()
{

int m;
m=2*f(3,g(4,5));
printf("\n m is %d",m);
}

what's the value of m
a.70
b.50
c.26
d. 69

8.
main()

```
{
char a[10];
strcpy(a,"\0");
if (a==NULL)
printf("\a is null");
else
printf("\n a is not null");}
```

what happens with it .
a. compile time error.
b. run-time error.
c. a is null
d. a is not null.

9. char a[5]="hello"

a. in array we can't do the operation .
b. size of a is too large
c. size of a is too small
d. nothing wrong with it .

10. local variables can be store by compiler
a. in register or heap
b. in register or stack
c .in stack or heap .
d. global memory.

11. average and worst time complexity in a sorted binary tree is

12. a tree is given and ask to find its meaning (parse-tree)
(_expression tree)
ans. ((a+b)-(c*d)) (not confirmed)
13. convert 40.xxxx into binary .

14. global variable conflicts due to multiple file occurance
is resolved during
a. compile-time
b. run-time
c. link-time
d. load-time

15.
two program is given of factorial.
one with recursion and one without recursion .
question was which program won't run for very big no. input because
of stack overfow .
a. i only (ans.)
b. ii only
c. i& ii both .
c. none

16.
```
struct a
{
int a;
char b;
int c;
```

```
}

union b
{
char a;
int b;
int c;
};
```
which is correct .
a. size of a is always diff. form size of b.(ans.)
b. size of a is always same form size of b.
c. we can't say anything because of not-homogeneous (not in ordered)
d. size of a can be same if ...

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int b=5;
a=&b;
}

main()
{
```

```
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}
```

what's the output .

1.10,5
2,10,10
c.5,5
d. none

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bye..
p.sreenivasa rao

_____

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From:
PVSAK Viswanadham Add to Address Book
Subject:
TI
Organization:
Computer Science Dept., Indian Institute of Technology, Kharagpur
To:
bkup

for TI aptitude test consist of all pictorial questions. ie in each question he will give 8 diagrams and ask to find the 9'th diagram in that sequence.
You go through RS Agarwal. These aptitude questins are very easy. Just pratice them. In RS Agarwal gothrough SERIES chapter. It is suffient. There are 35 aptitude questions. First 25 are very easy. Do these questions in just 15 or 20 minutes. Because last questions are very touch.

TECHNICAL TEST:
---

1)3 flipflops are connected so that after 0 to 5 count occured next number is zero. So what is the counter?
Ans: mod 6 counter

2)simplication of some boolean _expression which is simple.
Boolean _Expression is A+A'B.
Ans:A+B

3)Given inorder sequence and preorder sequence and asked to find out postorder sequence.

4)Some question on value of a static variable.

5) Given an interger in binary form,find the number of ones in that number without counting each bit.(This questin is not multiple choice question. This question carries more marks. So please take care for this question.)

6) 1-way set associative memory is called-----
a)direct b)something c)1-way set associative 4)something
Ans: c

7)Fastest IPC mechanism is

a)shared memory b)pipes c)named pipes d)semaphores
Ans:c

8)Some page references are given. You are asked to implement it with Least Frequently Used algorithm.

9)Some diagram is given.
Iam describinmg the diagram. A 2*1 MUX is given. The inputs are A,B. Output is C. C and A are tied together. What is the diagram.?

Ans:Latch.

****

This paper is for Electrical & Electronics students. There is separate test for computer Science Students. There are 20 questions.

1)Some circuit is given. Iam describing the circuit.
A resistor R & a capacitor C are connected in parallel.
To this circuit another circuit which is having a capacitor of capacity 2C & an impedence Z, is connected in series.
You are asked to find out the value of Z? Note that 2C & Z are connected in series.
a)Z=2C
b)Z=2L
c)Z=L/2
d)Z=2R

2)Some circuit which consist of only resistors R is given.
This is a repetative circuit. U have to find the effctive resistance of the entire circuit.
A)Rin=R
B)Rin=(5+sqrt(3))/7
C)Rin=(19+sqrt(3))/8
D)None.

3)Two wave forms are given. You are asked to write the cirsuit to get B(second wave form) from A(first wave form).

4)
```
#define SUM(a,b) a+b

main()
{
a=2;
b=3;
x=SUM(a,b)*2;
printf("x=%d\n",x);
}
```
Ans:8.

5)
```
number(int i)
{
number++;
printf("%d\n",number);
}

main()
```

```
{
static int i=0;
number(i);
}
```
Ans: I don't know.

6)Some circuit is given. I can't describe the circuit. There are
3 resistors,3 capacitors & one inverter.. The question is
What is the value of the frequency such that the circuit oscillates.
A)f=RC
B)f=sqrt(3)/(Pi*R*C)
C)f=1/(Pi*R*C)
D)something
Ans:I don't know the answer.

7)Question on flipflop. So gothrough all flipflops.

8)There are 5 questions on Nmos & Pmos circuits.

****
This Paper is for Computer Science Students. THis paper is
very easy. You can definitely do it in one hour.
****

(1) The fastest memory is
(i) DRAM, (ii) ROM, (iii) SRAM, (iv) Main memory
Ans : SRAM

(2) Programing exceptions are
(i) Asynchronous, (ii) Synchronous, (iii) None
Ans : Asynchronous

(3) DSP which architecture is used
(i) MIMD, (ii) SIMD, (iii) Nueman, (iv) Harvard Architecture
Ans : Harvard Architecture

(4) C prog. for searching for an element in linked list

(5)
```
main()
{
unsigned char i;
int sum;

for(i=0; i<300; i++)
sum+ = i;
printf("\nSum = %d\n", sum);
}
```

Ans : infinite loop

(6)
```
void fn(int *p)
{
static int val = 100;
p = &val;
}

main()
```

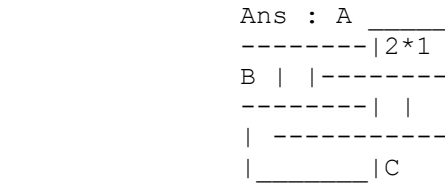
```
{
int i=10;
printf("i=%d\n", i);
fn(&i);
printf("i=%d\n", i);
}
```

Ans : i=10 i=10

(7) int a[10[15];
char b[10[15];
(a) location g a[3][4], if base location g a[0][0] is ox1000
(b) location g b[3][4], if base location g b[0][0] is ox2000
int taken 32 bits and char taken 8 bits.

Ans : (a) ox10C4 (b) ox2031

(8) Implement OR gate function with 2*1 MUX

Ans : A


B=C

(9) Implement 4*1 MUX with 2*1 MUXES

(10) Swapping without using a temporary variables. (2 methods)

(i) x = x+y;
y = x-y;
x = x-y;

(ii) x = x^y;
y = x^y;
x = x^y;

(11) Count no of 1's in a word without using bit by bit.
(This question carries more marks. It is not a multiple choice question.)

(12) Code 1 :
for(i=0; i<1000; i++)
for(j=0; j<100; j++)
x = y;

Code 2 :
for(i=0; i<100; i++)
for(j=0; j<1000; j++)
x = y;

Which code will execute faster
(i) Code 1 and Code 2 are of same speed,

(ii) Code 1,
(iii) Code 2,
(iv) None.

Ans : Code 2

(13) main()
{
int a[10] = {1, 2, 3, ..., 10}, i, x=10, temp;

for(i=0; i temp = a[i];
a[i] = a[x-i-1];
a[x-i-1] = temp;
}
(i) All contents of array a are reversed
(ii) Only some portions are altered
(iii) Remains same
(iv) None

Ans : (iii)

(14) An array is stored in row major order. The memory capacity is 30 MB. And in unix system demand paging is used. Which one will give more page faults?

#define V_L_I 10000
int i, j, array[V_L_I][V_L_I];

Code 1 :
array[i][j] = 1;

Code 1 :
for(j=0; j for(i=0; i array[i][j] = 1;

Ans : Code 2

(15) In C which parameter passing technique is used?
(i) call by value,
(ii) call by reference,
(iii) both

Ans : call by value

(16) A circuit is given with 2 exclusive OR gates whose boolean _expression will be y = '(AB) + AB
(' indicates bar)

(17) main()
{
int i = 1;
fork();
fork();
printf("\ni = %d\n", i+1);
}

Ans : 4 printfs will occur and i = 2

(18) Compute the complexity of Binary search.
Ans : O(lg n) (Answer in detail. This is not a multiple choice question.
It carries more marks.)

(19) Write _expression for the tree graph :
Ans : ((a-b) + c*d)/x

(20) # define MAX(a, b) a>b ? a:b
```
main()
{
int m, n;
m = 3 + MAX(2, 3);
n = 2 * MAX(3, 2);
printf("m = %d, n = %d\n", m, n)
}
```

Ans : m=2, n=3
paper of texas instruments.
Technical + aptitude + interview.
* Questions on c
here i am not strictly following syntax it is just to show what was asked.
1 #define sum(a,b) a+b
value of sum(2,3)*2 Ans:8

2. a=5,b=6
swap(&a,b);
This function is written to swap a and b
find value of a and b . Ans 6,6

3.
```
function()
{
static int a=0;
a=a+1;
return a;
}
main()
{
function();
function();
function();
printf a;
}
```
final value of a ? Ans : a=3. static initializes once.
4.Write two prog. to swap a & b without using temp variable.
5.
```
unsigned char i;
int sum=0;
for(i=0;i<300;i++)
sum=sum+i;
printf(sum);
```

Ans:Program will held in infinite loop b/c i can not exceed 255.

6.

five questions on MOSFETS.
four were having single mosfets.
questions were simple.as i told u in Gwalior.
just on the funda that it will conduct if Vg-Vs > Vt .

In one question output at drain was to be calculated while o/p was initially
charged to 5v and to the gate 5v were applied.

In one question output at drain was to be calculated while o/p was initially
charged to 5v and the gate was shorted to drain.
7.
Clear the concept of settling time , hold time and other times. 3 ques on that.
like values of various delays were given and max frequency at which the circuit can work
hint : 1/sum of all delays . In our case ans was 200 Mhz.

8. An input and output waveform was given and circuit was to be designed with the use of one
delay.
Ans : exor gate in which second input is first input with a delay.

9. A question to determine sequence of counter. Don't get puzzled it was a
tough question.

10.The output and input of a inverter is connected by three RC stages in between of each stage two amplifiers with poles at imaginary axis were connected.
Hint : The poles at imaginary axis will create extra 180 phase shift thus the
circuit will oscillate and calculate the frequency of operation.

11. A series of infinite connected rc circuit and overall input resistance is
calculated.(question of 12 class)

1: given an expression tree and asked us to write the in fix of that expression

four choices

2:
global variables in different files are

a:at compiletime
b) loading time
c) linking time
d)execution time

3)size of(int)
a) always 2 bytes
b) depends on compiler that is being used
c) always 32 bits

d) can't tell

4)which one will over flow given two programs
2

prog 1:

```
main()
{
int fact;
long int x;
fact=factoral(x);

}
```

prog2:

```
main()
{
        int fact=0
for(i=1;i<=n;i++)
        fact=fact*i;

}
```

```
int factorial(long int x)
{

if(x>1) return(x*factorial(x-1);
}
```

a) program 1;
b) program 2;
c) both 1 &2
d) none

}

5) variables of fuction call are allocated in
a) registers and stack
b) registers and heap
c) stack and heap
d)

6)

avg and worst case time of sorted binary tree

7) data structure used for proority queue
a) linked list b) double linkedd list c)array d) tree

8)

```
 main(){
char str[5]="hello";
if(str==NULL) printf("string null");
else printf("string not null");
}
```

what is out put of the program?
a) string is null b) string is not null c) error in program d) it executes but print nothing

9)there are 0ne 5 pipe line and another 12 pipe line sates are there and flushed time taken to execute five instructions

a) 10,17
b) 9,16
c)25,144
d)

10)

for hashing which is best on terms of buckets
a)100 b)50 c)21 d)32 ans 32

11)

```
void f(int value){
for (i=0;i<16;i++){
if(value &0x8000>>1) printf("1")
else printf("0");
}
}
```

what is printed?
a) bineray value of argument b)bcd value c) hex value d) octal value

12)

```
void f(int *p){
static val=100;
val=&p;
}
main(){
int a=10;
printf("%d ",a);
f(&a);
printf("%d ",a);
}
```

what will be out put?
a)10,10

13)

```
struck a{
int x;
float y;
char c[10];
}
union b{
int x;
float y;
char c[10];
}
```

which is true?
a) size of(a)!=sizeof(b);
b)

c)
d)

14)

```
# define f(a,b) a+b
#defiune g(c,d) c*d
```

find valueof f(4,g(5,6))
a)26 b)51 c) d)

15)

find avg access time of cache
a)tc*h+(1-h)*tm b)tcH+tmH

c) d) tc is time to access cache tm is time to access when miss occure

16)

```
main()
{
char a[10]="hello";
strcpy(a,'\0');
printf("%s",a);
}
```

out put of the program?
a) string is null b) string is not null c) program error d)

17)

simplyfy k map

1 x x 0
1 x 0 1

18)

```
int f(int a)
{
a=+b;

//some stuff


}

main()
{
x=fn(a);
y=&fn;
```

what are x & y types
a) x is int y is pointer to afunction which takes integer value

19) char a[5][15];
int b[5][15];
address of a 0x1000 and b is 0x2000 find address of a[3][4] and b[3][4]
assume char is 8 bits and int is 32 bits

a) b) c) d)

there are 20 questions all in techinical paper and 36 questions in appititude test

in appititude thay have given all diagrams and asked to find what comes next

thay are quite easy and i hope if u practice r.s aggraval u can do it easily

for tecnical thay have given 1 hr for 20 questions and for not technical thay have given only 40 min and 36 questions

this is the paper i have right now

Here is Texas paper for you.
in this paper there was 20 questions as follows in 60 minutes .
second part consists of 36 que. in 30 minutes all questions are diagramatical.(figurs)..

1. if a 5-stage pipe-line is flushed and then we have to execute 5 and
12
instructions respectively then no. of cycles will be
a. 5 and 12
b. 6 and 13
c. 9 and 16
d.none

2. k-map

```
ab
----------
c       1   x  0  0
       1   x  0  x
```

solve it

a. A.B
B. ~A
C. ~B
D. A+B

3.CHAR A[10][15] AND INT B[10][15] IS DEFINED
WHAT'S THE ADDRESS OF A[3][4] AND B[3][4]
IF ADDRESS OD A IS OX1000 AND B IS 0X2000

A. 0X1030 AND 0X20C3

B. OX1031 AND OX20C4
AND SOME OTHERS..

```
4. int f(int *a)
{
int b=5;
a=&b;
}

main()
{
int i;
printf("\n %d",i);
f(&i);
printf("\n %d",i);
}
```

what's the output .

1.10,5
2,10,10
c.5,5
d. none

```
5. main()
{
int i;
fork();
fork();
fork();
printf("----");
}
```

how many times the printf will be executed .
a.3
b. 6
c.5
d. 8

```
6.
void f(int i)
{
int j;
for (j=0;j<16;j++)
{
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
```
what's the purpose of the program

a. its output is hex representation of i
b. bcd
c. binary
d. decimal

7.
```
#define f(a,b) a+b
#define g(a,b) a*b

main()
{

int m;
m=2*f(3,g(4,5));
printf("\n m is %d",m);
}
```

what's the value of m
a.70
b.50
c.26
d. 69

8.
```
main()
{
char a[10];
strcpy(a,"\0");
if (a==NULL)
printf("\a is null");
else
printf("\n a is not null");}
```

what happens with it .
a. compile time error.
b. run-time error.
c. a is null
d. a is not null.

9. `char a[5]="hello"`

a. in array we can't do the operation .
b. size of a is too large
c. size of a is too small
d. nothing wrong with it .

10. local variables can be store by compiler
a. in register or heap
b. in register or stack
c .in stack or heap .
d. global memory.

11. average and worst time complexity in a sorted binary tree is

12. a tree is given and ask to find its meaning (parse-tree)
(expression tree)
ans. ((a+b)-(c*d)) (not confirmed)
13. convert 40.xxxx into binary .

14. global variable conflicts due to multiple file occurance
is resolved during
a. compile-time

b. run-time
c. link-time
d. load-time

15.
two program is given of factorial.
one with recursion and one without recursion .
question was which program won't run for very big no. input
because
of stack overfow .
a. i only (ans.)
b. ii only
c. i& ii both .
c. none

16.
struct a
{
int a;
char b;
int c;
}

union b
{
char a;
int b;
int c;
};
which is correct .
a. size of a is always diff. form size of b.(ans.)
b. size of a is always same form size of b.
c. we can't say anything because of not-homogeneous (not in ordered)
d. size of a can be same if ...

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	ab			
c	1	x	0	0
	1	x	0	x

solve it

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```
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void f(int i)
{
int j;
for (j=0;j<16;j++)
{
```

```
if (i & (0x8000>>j))
printf("1");
else
printf("0");
}
}
```
what's the purpose of the program

a. its output is hex representation of i
b. bcd
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```

what's the value of m
a.70
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d. 69

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else
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b. run-time error.
c. a is null
d. a is not null.

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```

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a. in register or heap
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c .in stack or heap .

```
    d. global memory.

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    12. a tree is given and ask to find its meaning (parse-tree)
    (expression tree)
    ans. ((a+b)-(c*d))  ( not confirmed)
    13. convert 40.xxxx into binary .

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    is resolved during
    a. compile-time
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    one  with  recursion and one without recursion .
    question was which program won't run for very big no. input
because
of stack overfow .
    a. i only  (ans.)
    b. ii only
    c. i& ii both .
    c. none

    16.
    struct a
    {
    int a;
    char b;
    int c;
    }

    union b
    {
    char a;
    int b;
    int c;
    };
    which is correct .
    a. size of a is always diff. form size of b.(ans.)
    b. size of a is always same  form size of b.
    c. we can't say anything because of not-homogeneous (not in ordered)
    d. size of a can be same if ...
```

Composed by Ram:

DIGITAL DESIGN

1.Using a 2:1 Mux realize the following

a) NOT gate b) AND gate c) OR gate d) Ex-OR gate e) Ex-NOR gate
f) NAND gate g) NOR gate h) Latch i) FlipFlop

Answer:
For these kind of questions always use Shannon's Expansion.
hint : Use Shannon's Expansion , get expression in the form of Mux equation
muxout = sel_bar * Input0 + sel*Input1.

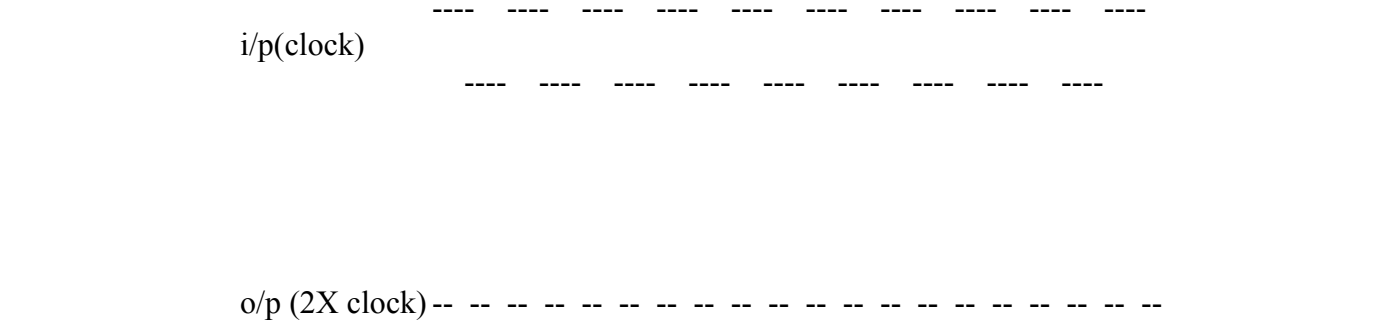
Ex: Realize a 2-i/p AND gate using a 2:1 mux.

AND gate: Y = A*B.
= A*B + ~A*'0'

Now select A as Mux control signal and Input0 is '0' (ground potential/electrical equivalent of logic '0').
Input1 is 'B'.

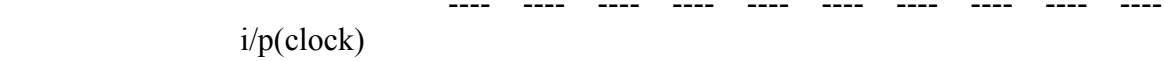
2.Using Combo logic Multiply Clock by two (freq of clock at o/p = 2* freq at i/p).

Answer:
For these kind of questions, first draw the i/p and o/p waveforms, then try to add one or more waveforms which applied to a gate (or a combination of gates) will give the o/p waveform.



Now try to find a gate and an i/p x which when applied along with the i/p clock to the gate (combo gate cluster)

this is purely based on systematic approach... develop it...
you should be able to find that if the i/p clock is delayed by T/4 (where T is the period of the clock) and this applied to Ex-OR gate along with the actual clock would give the 2xclock.
Dont worry about the delay element for T/4, that would not be difficult, you can add a buffer.
Now try to get 3X clock using combo logic only. (you may need more than two i/ps ;)).





3.Realize a transistor level circuit for
Y = { [(ABC+Abar)bar] * (AB + Bbar) }

4.Given/using a Positive Trigger as input generate Square wave.

5. Question on Static Hazards
AND gate 1 has two i/ps A , sel
AND gate 2 has two i/ps B,sel_bar
output of these AND gates are given as i/p to ex-or gate
Tand = Tex-or= 2ns, Tinv (used for sel_bar) = 1ns

find Glitch width and draw the hazard-free circuit
hint: See switching theory book by Kohavi

6.Draw FSM for "0101" sequence detector and code it in Verilog/VHDL.
How many FFs are needed?

7.Given a 8 bit number how would you check whether it is a palindrome or not???

8.Two FFs are cascaded with combo logic in between (Q of FF1 to D of FF2)
Tcombo,min = 1ns and Tcombo,max = 3ns
Tsetup = Thold = 2ns, Tclk = 10ns, Tclock-to-Q = 2ns
check for Setup and hold time violations.

9.What is Synchronizer used for ? draw the ciruit and comment on sizing of Txs.
hint : see DIGITAL INTEGRATED CIRCUITS book by Rabaey.

10.Draw Tx level ckt for Y= AB + AC + BD + CD.

11.What is RACE condition ? How to avoid it?

12.Using D FF and combo logic realize T FF.

13.Using D FF and COMBO logic realize JK FF.

14.What are the advantages and disadvanteages of Dynamic Logic ?

15.Draw NAND and NOR equivalents of CMOS inverter for equal rise and fall times.
hint : see DIGITAL INTEGRATED CIRCUITS by Rabaey.

16. Is it possible to have negative setup and hold times ? Explain.

17.A 7 bit ring counter has initial state 0100010 after how many clock cycles it will return to initial state?

18. Which device is fast BJT or MOS? Why ?

19. A 4 bit shift register has _____ number of states.

20. What is Mealy FSM and Moore FSM? Which one is fast?

21.Give adv and disadv of Mealy and Moore FSMs? Give examples of applications of both.

22.Swap two 8-bit registers without using another register.
hint : use boolean logic

23.Realize a two i/p AND gate using Ex-OR gate .
hint: don't waste time , come 2 a conclusion , ... ya u r right.... :-)

24.Describe an FSM to detect three successive coin tosses that result in Heads.

25.In what cases do you need to double clock a signal before presneting it to a Synchronous state machine?

26.You have a driver that drives a long signal and connects to an i/p device. At the i/p there is either overshoot or undershoot or signal threshold violations. What can be done to correct this problem?

27.What is a Silicon Compiler and a Memory Compiler used for?

28.To realize a 4x4 multiplier using ROM, what is the size of ROM needed?

29.In a system there are two modules A and B. A is operating at 25 MHz and B at 25 KHz From module A if a pulse of width equal to width of clock (1/25 Micro seconds) is sent, How ensure that the pulse will be correctly received at module B without using handshaking or Buffers like FIFO?

30.A D FF has its D i/p from a MUX. MUX input0 is connected to external i/p and MUXi input1 is connected to output of D FF (Q) through combo block(i.e: feedback of o/p to

i/p thru combo block). If Mux delay is 0 ns and Tsetup = 3ns, Thold = 2ns , TClock-to-Q = 1ns What is the max frequency of the circuit with and without feedbak?

31.Why PMOS Tx is made 2.5 times wider than NMOS ?

32.If PMOS and NMOS Txs are interchanged in a CMOS inverter, what does it work like?

33.Draw Ids-Vds curve of a MOSFET with
a)increasing VGS, b) increasing W, c) considering Channel Length modulation

34.Why MOSFET goes into saturation and what type of current flows (drift/diffusion) at saturation?
(or)
If channel is pinched of how current flows from source to drain ?

35.List variuos Capacitances in a MOS device and their approximate values in Linear , saturaiton and cut-off regions.

36.Explain VTC of a CMOS inverter .what is the effect of channel length modulation in VTC ?

37.How to increase gain of a CMOS inverter in transition region ?On what factors does it depend?

38. What is Noise Margin, Noise Immunity? differentiate.

39.What is regenerative property of a CMOS inverter? explain with graphs.

40.What is Switching/logic threshold of a CMOS inverter ? How to change it?

41.How to measure Noise Margin?

42.What is Body effect?

43.What is CMOS latchup ? how to avoid it?

44.What is Electromigration ? How to avoid it ?

45.What is ESD ? How to avoid it?

46.What is Ground Bounce ? How to avoid it?

47.Why don't you use a NMOS/PMOS as a TG?

48.What is Full scaling and constant voltage scaling ?

49.Why scaling is done?

50. If a technology is scaled by 30 % (VDD also), how the following change
a) Cox,Cg b) Power c) Area d) Delay.

51.GIve the Expression for Elmore delay and penfield Rubenstein delay models.

52.Why NAND logic is preferred in CMOS ?

53.What happpens if we increase number of contacts and vias from one metal layer to another?

54.Draw a 2 i/p NAND gate and explain sizing regarding Vth and rise/fall times.

55.What are limitations in increasing Vdd to reduce intrinsic dcelay?

56.What happens to delay if we include a resistence at the o/p of a cmos ckt?

57.What is crosstalk ? On what factors does it depend?

58.What are various kinds of power dissipation in CMOS circuits?

59.What are the disadvantages of scaling?

60.You have three adjacent parallel metal lines.Two out of phase signals pass through outer
lines.Draw the signal in central metal line due to interference. repeat for inphase signals in the outer lines.

61.What happens if we increase no: of contacts or vias from one metal layer to another?

62.Draw Tx level ckt for a 2-i/p NAND gate and explain sizing considering
a) Logic threshold b) equal rise and fall times.

63. Why is it preferred to have logic threshold at Vdd/2 ?

64.What is Self-loading ?

65.Let A and B are inputs to a two i/p NAND gate, which signal should be close to the output
a) if signal A arrives later than signal B,
b) if signal B has higher switching activity than signal A,

66.Why fan-in of gates is resricted to 4 ?What is done to have large fan-in ?

67.Draw Stick diagram of a NOR gate and optimize it.

68.Give various methods used for reducing power in CMOS ciruits.

69.What is charge sharing ? Explain charge sharing while sampling data from a bus.

70.When driving a large capacitive load why do we use a chain of inverters with progressive
increase in size, instead of having a large buffer?

71.Explain difference between normal Buffers and Clock buffers.

72.Mention algorithms used for CLOCK distribution.

73.While laying out a large(wide) Transistor , why do we connect small transistors in parallel
rather than laying out a Tx with large width?

74.Why don't we use NMOS or PMOS as a switch?

75.Draw 6T SRAM cell . Explain read and write operation.
which one takes more time read/write ? why?

76.Draw a Differntial Sense amplifier and expalin its operation.

77.Draw a Cross coupled Snese amp and expalin its operation.

78.What is a double stage Differential sense Amplifier? what is it needed for?

79.Comment on sizing of Access Tx used in 6T SRAM cell.

80.Which one is fast NAND/ NOR ROM ?Give applications of each?

81.In memory design interconnect delay becomes critical , How is it reduced?

82.How does size of a PMOS pull up Tx affect performance of a 6T SRAM cell?

83.Explain sizing of variuos Txs used in SRAM cell.

84.What is critical path in SRAM?

85.In SRAM which metal layers would you prefer for word and bit lines?why?

86.How do you model SRAM in RTL ?

87.For an AND-OR implementation of a 2:1 Mux, how would you check for stuck-at-faults at internal nodes?

88. Mention algorithms used for Stuck-at-fault analysis.

89.What is the differnce between testing and verification?

90.What Kind of circuit is this
A and B are inputs to an AND gate
AND gate output goes to one i/p of OR gate
The other i/p of OR gate comes from a Ex-OR gate
inputs to the Ex-OR gate are C and the output of the OR gate
(final output fedback to i/p)
combo/sequential?
synchronous/asynchronous?

91.Realize the boolean function
Y= A'B'C +A'BC+ABC+ABC'+AB'C
a) using 2-i/p and 3-i/p NAND gate,
b) using 2-i/p and 3-i/p NOR gate
c) using AOI gate
d) using inverter

92.What is the importance of SCAN in a digital system?

93. A Ex-OR B = C, Prove that
a) B Ex-OR C = A,
b) A Ex-OR B Ex-OR C = 0.

94.Construct a test pattern that can detect stuck-at-1 fault in the ckt given below
NAND gate NAND1 has two i/ps C and D
NAND gate NAND2 has two i/ps A and Y
AND gate has o/ps of NAND gates NAND1 and NAND2 as i/ps
and its o/p is Y (this is fedback to i/p of NAND gate NAND2)

95.In an Op-Amp ckt i/p offest is 5mv, Voltage gain = 10,000, Vsat = +/- 15v.Find o/p voltage.

96.Draw P-n/w for the function Y = ((AB+C) D)'.

97.Realize JK FF using D FF and MUX.

98.Realize the function Y= A + BC' + BC (A + B) using 2:1 Mux.

99.For the circuit given below
D FF "DFF1" has its D i/p,D1, connected to o/p of Ex-OR "Ex-OR1"gate.

D FF "DFF2" has its D i/p,D2, connected to o/p of Ex-OR gate "Ex-OR1".
i/ps of Ex-OR gate "Ex-OR1" are o/ps of "DFF1" and "DFF2" (Q1 and Q2)
CLK i/p of "DFF1" is connected directly to clock signal and CLK i/p of "DFF2"
is connected to inverted clock signal (clcok signal goes to DFF2 thru inverter).
What is the realtion between input and output frequencies?

100.Design a Synchronous ckt for the following clock waveform
CLK ---> thrice the CLK period ---> half the period of i/p

101.What are setup and hold times of a FF? What happens if we don't consider them when
designing a digital circuit?

102.Two D FFs, "DFF1" and "DFF2" are cascaded, if Tsetup = Thold = 2ns and Twire = 0ns.What is the max Clock frequency for the ckt ? If DFF2 is negative edge triggered D FF then what is the maximum clock frequency?

103.What is a FIFO buffer ? What is a FIFO buffer used for ?Give example.

104.How can you make sure that Glitches does not occur in a circuit at logic level?

105.What is the function of a D FF whose Complemented o/p (Qbar) is connected to it's input,D. What is the max clock frequency that can be used for it?

106.What happens if Setup violation occurs ? what happens if Hold violation occurs? Can a circuit have both setup and hold violations? Is it possible to have Setup and hold violations together on the same path?

107. Which one will have less switching activity ?
a) Tree real;ization or b) chain realization .

108.Two D FFs,DFF1 and DFF2 are cscaded and clock arrives late at the clcok input of DFF2.
What happens if the delay (in path from clock signal to clk i/p of DFF2) is large?How can this problem be solved?

109.Design a divide-by-3 sequential circuit with 50% duty cycle.

110.Draw the circuit of a TG based Latch.

111. _____

i/p ----------Buffer----------o/p
In the above circuit, what is the purpose of the buffer.(Note that o/p is fedback to i/p)?
Is it redundant /necessary to have a buffer?

112.What is the o/p of the ciruit given below
2-i/p Ex-OR "Ex-OR1" has its i/ps tied to X,
2-i/p Ex-OR "Ex-OR2" has one of it's i/p connected to o/p of "Ex-OR1"
and the other i/p connected to X.
2-i/p Ex-OR "Ex-OR3" has one of it's i/p connected to o/p of "Ex-OR2"
and the other i/p connected to X.
What is the o/p of the circuit(o/p of "Ex-OR3").

113.Given a Circular disk with a sector of 45 degrees painted in blue. Two sensors are given and they can detect change in color. Design a circuit with minimum number of gates to detect the direction of the disk when it is rotated.

114.Given two transparent latches, realize a positive edge triggered D FF using minimum number of gates.

115.How many 2:1 Muxes are needed to realize a 16:1 Mux?

116.What is metastability? Why it occurs ? How to avoid it?

117.Convert a 2-i/p NAND gate to an inverter in two different ways.

118.Realize a T FF using 2:1 Muxes and few gates.

119.Realize D FF from RS latch (not Flip Flop).

120.What is the difference between EEPROM and Flash Memory?

121.Define Clock skew. What are the causes for it ? How Positive skew effects the system?

122.Define Clock jitter and differentiate skew and jitter.How clock jitter effects the system?

123.Which one is good Synchronous reset or Asynchronous reset?

124.Describe an FSM to detect the string "abca" if i/ps are a,b,c,d. Code it in verilog/VHDL.

125.Change rise and fall times of a CMOS inverter without changing W/L ratios.
hint: rise and fall time depend on current drive available.

126.What are setup and hold times? what do they signify ? which one is critical for estimating maximum clock frequency?

127.Suppose you have a combo ckt b/w two registers driven by a clock.If the delay of Combo ckt is larger than the clock period, then how would you overcome the problem?

128.The answer to the above question is break the combo ckt (functionality of combo into simple functions) and pipeline the combo block.What is the penalty in doing so?

129.Draw the ckts of TG based D latch and D FlipFlop(positive edge triggered). how would you reduce load on the clock signal? what is the penalty in doing so?

130.Realize Ex-OR using TGs and modify to Ex-NOR gate (without complementing o/p).

131.Design an FSM to give modulo-3 counter when input X=0 and modulo-4 counter when input X=1.

132.What is clock feedthrough?

133.Given a Clock signal, generate nonoverlapping clcoks (clock and clock_bar) using Combo logic.

134. What happens to VTC of a CMOS inverter, if supply voltage is reduced?

135.What are the limitations on reducing Vdd from delay point of view and from noise point of view?

136.Design a logic circuit using AOI configuration sich that if input a=1, output Y = AB+CD else Y=DE + CF.

137.What is charge sharing? how to avoid it?

138.Design a ckt that clips every alternate clock pulse.

139.If A ? B = C and A?C = B, then what is the operator "?".

140.Dynamic circuits with feedback are called _____?

141.Design a circuit to count No: of ones in a 7-bit binary number (data comes in parallel). (do not do it bit by bit)

142.Generate a square wave using Mux.

143.Draw CMOS ckt for a Tri-state Buffer.Realize a 2:1 Mux using Tri-state Buffer.
COMPUTER ORGANIZATION:

Hi folks,
I thought, Computer organization is required for a VLSI design engineer.Intel,amd,....do processor design and expect you to have "what is what" knowledge, you may not be doing the architecture development but nothing wrong in knowing "what is what ".....

these are the Questions I have collected from my frens (and personal experience).

1.What is a Cache? What is it used for? What is the principle behind it?

2.what should be the size of a cache -- large/small?

3. What is a cache hit and cache hit ratio?

4. what are the various mappings used in Cache?
(direct, assosciative , set-assosciative)

5.What are the stages of a 5 stage DLX pipeline?

6. What are bubbles in a pipeline ?

7. What are HAZARDS in a pipelined system?

8. What is the ideal throughput of a N stage pipeline system? What prevents from achieving the
ideal throughput ? Is it better to have a 5 stage pipeline or 20 stage pipeline?

9.Expand TLB. what is it used for?

10. Name some Bus standards u know. Compare them.

11.Explain purpose of cache in a single Processor system and a double processor system with a
separate cache for each processor.

12.Explain difference between "Write through" and "Write back" caches.

13.What is MESI ?

14.What is Snooping?

15.Swap two 8-bit registers without using any other register.

16.Differentiate Overflow and Carry flag.

17.Differntiate Superscalar and VLIW processors.

18.What is MicroProgram control and Hardwired control?

19.What is Von-Numan architecture and Harvard architecture ?

Which one is used for MicroProcessor and which one forDigital signal Processor? Why?

20.What is Branch Prediction and BTB?

21.What is virtual memory?

22.What is cache Cohorency?

23.Differntiate MicroProcessor and MicroController.
Ans: In addition to all arithmetic and logic elements of a general purpose microprocessor, the microcontroller usually also integrates additional elements such as read-only and read-write memory, and input/output interfaces.

24.Processor is busy , but you want to perform some task . How will you do that?
Ans: Interrupts (Interrupts are used to pause execution of processor's program service a routine and then continue with the program)

25.What is ACBF (hex number) divided by 16 , give Quotient and remainder?

26.Given cache size is 64KB , Block size is 32B and the cache is two-way set assosciative.
For a 32-bit physical address, give the division between block offset, index and tag.

27.Differentiate RISC and CISC. Is RISC always fast?

28. How is a DSP different from a GPP?
Ans:The essential difference between a DSP and a microprocessor is that a DSP processor has features designed to support high-performance, repetitive, numerically intensive tasks. In contrast, general-purpose processors or microcontrollers (GPPs/MCUs for short) are either not specialized for a specific kind of applications (in the case of general-purpose processors), or they are designed for control-oriented applications (in the case of microcontrollers). Features that accelerate performance in DSP applications include:

* Single-cycle multiply-accumulate capability; high-performance DSPs often have two multipliers that enable two multiply-accumulate operations per instruction cycle; some DSP have four or more multipliers
* Specialized addressing modes, for example, pre- and post-modification of address pointers, circular addressing, and bit-reversed addressing
* Most DSPs provide various configurations of on-chip memory and peripherals tailored for DSP applications. DSPs generally feature multiple-access memory architectures that enable DSPs to complete several accesses to memory in a single instruction cycle

* Specialized execution control. Usually, DSP processors provide a loop instruction that allows tight loops to be repeated without spending any instruction cycles for updating and testing the loop counter or for jumping back to the top of the loop

* DSP processors are known for their irregular instruction sets, which generally allow several operations to be encoded in a single instruction. For example, a processor that uses 32-bit instructions may encode two additions, two multiplications, and four 16-bit data moves into a single instruction. In general, DSP processor instruction sets allow a data move to be performed in parallel with an arithmetic operation. GPPs/MCUs, in contrast, usually specify a single operation per instruction

While the above differences traditionally distinguish DSPs from GPPs/MCUs, in practice it is not important what kind of processor you choose. What is really important is to choose the processor that is best suited for your application; if a GPP/MCU is better suited for your DSP application than a DSP processor, the processor of choice is the GPP/MCU. It is also worth noting that the difference between DSPs and GPPs/MCUs is fading: many GPPs/MCUs now include DSP features, and DSPs are increasingly adding microcontroller features.